SFW / CRM ADD-ON + STANDALONE SAAS + RESELLER READY

AI Revenue & Omnichannel Marketing Automation SaaS

Developer Master Blueprint • Multi-Tenant • WhatsApp + RCS + SMS • AI Chatbot • Self-Marketing • CRM-Agnostic

Production implementation guide for building, launching and scaling a highly automated AI revenue, chatbot and omnichannel marketing operating system

Version 2.0
Prepared: 27 August 2026
Target: Standalone SaaS + SFW Add-on + External CRM + White-label / Reseller
Channels: WhatsApp Business Platform + RCS + SMS + Email-ready

BUILD PRINCIPLE: Launch a sellable modular monolith first. Preserve tenant isolation, events, adapters, auditability and idempotency from day one. Split services only when scale metrics justify it.

Document Purpose

This document is the authoritative engineering blueprint for an AI-driven, multi-tenant revenue and marketing automation platform that can be sold as a standalone SaaS, embedded as an add-on inside Smart Field Work (SFW), white-labeled/resold, or connected to third-party CRMs. Version 2.0 extends the original Revenue OS with one-click WhatsApp onboarding, an AI/no-code chatbot builder, AI-assisted template approval workflows, bulk campaign orchestration, RCS and India-compliant SMS adapters, self-marketing autopilot, reseller wallets, rate-card billing, and cross-channel revenue optimization. It is implementation-oriented: a capable Laravel/React/Flutter team should be able to convert the specification directly into epics, migrations, APIs, queues, UI screens and automated tests.

Success is measured by safe, attributable revenue outcomes—not by message volume. "110% target" is implemented as a stretch-goal controller and recovery engine, never as a guaranteed outcome.

Revision Control

Field	Value
Document owner	Product / Engineering
Version	1.0
Date	27 August 2026
Primary launch mode	SaaS + SFW add-on
Initial channel	WhatsApp Business Platform
Future channels	Email, SMS, voice, web chat, social messaging where permitted
Primary regions	India-first, international-ready
Status	Build-ready master specification

How Developers Should Use This Document

1. Create epics from each major module and preserve the numbered acceptance criteria.
2. Implement Phase 1 exactly as the "Fastest Production Path"; defer optional complexity until the stated scale trigger.
3. Keep every domain object tenant-scoped and every external action idempotent.
4. Use deterministic workflow/rule code for policy, limits, billing and state transitions; use AI only for tasks that benefit from probabilistic reasoning or language generation.
5. Instrument every automation so product management can measure funnel conversion, cost, latency, failure rate and revenue attribution.

Executive Summary

The product is an Autonomous Revenue OS: a multi-tenant platform that receives leads and customer events, verifies communication eligibility, scores opportunities, generates context-aware messages, runs follow-ups, books demos, assists negotiation, recovers payments, reactivates lost opportunities, updates CRMs and continuously recalculates the actions required to hit daily, weekly and monthly targets.

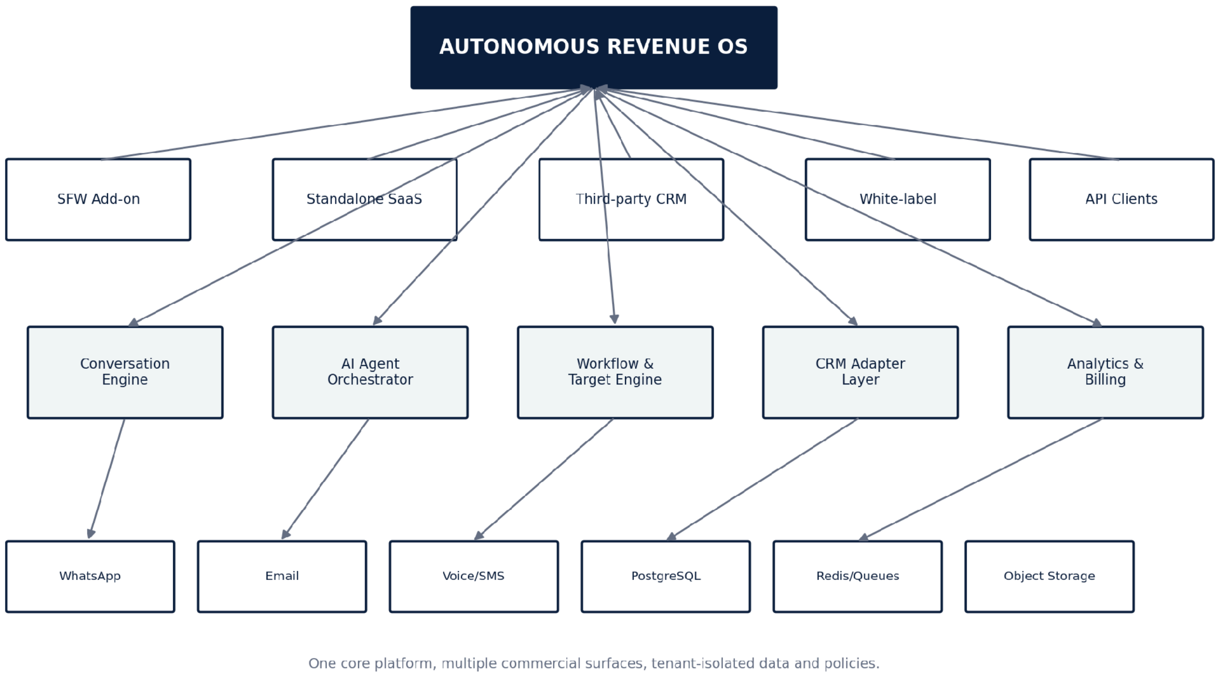

Figure 1 — One core platform, four commercial modes.

Commercial Modes

Mode	Customer experience	Integration
Standalone SaaS	Customer logs into your branded portal	Native tenant workspace
SFW Add-on	AI Sales appears inside SFW navigation	SSO + two-way events/API + embedded UI
Third-party CRM	Customer keeps existing CRM	Connector/adaptor + webhooks + field mapping
White-label / OEM	Partner brands the platform	Custom domain, theme, SSO, configurable feature set

Core Differentiators

- Revenue-target controller that converts goals into required pipeline activity and reprioritizes work during the day.
- AI/Human hybrid modes: Full AI, AI Assist and Human-led with suggested replies.
- Industry and company knowledge packs with RAG, product rules, pricing authority and objection libraries.
- Consent, frequency, template-window and opt-out policy gates before any outbound action.
- Universal CRM object model and adapter SDK so SFW is a first-class client rather than a hard-coded dependency.
- Event-driven execution, audit trails and cost metering suitable for both SMB and enterprise tenants.
- Continuous experimentation optimized for paid conversions and revenue, not vanity metrics.

Non-Goals for Version 1

- Do not build a generic all-purpose marketing automation suite before the sales funnel is proven.
- Do not scrape Google Maps or make prohibited data extraction a platform dependency.
- Do not create one infrastructure service per AI agent.
- Do not allow the LLM to decide legal eligibility, billing, discount limits or irreversible record changes without deterministic guards.
- Do not start with complex Kafka/service-mesh architecture unless traffic proves the need.

Current Platform Guardrails

WhatsApp Business messaging requires opt-in, approved templates for business-initiated messaging, a 24-hour customer-service window for free-form replies after the user message, clear escalation paths from automation, and compliance with applicable law. The product must therefore store consent evidence and run a policy gate before every outbound action.

India privacy design should align with the Digital Personal Data Protection Act, 2023 and the notified Digital Personal Data Protection Rules, 2025, including clear notice, purpose limitation, consent records, withdrawal handling and security safeguards. Compliance details must be reviewed by counsel for each target geography and vertical.

PART I — PRODUCT & ARCHITECTURE

What to build, how it is sold, and the fastest scalable technical shape.

1. Product Vision and Success Criteria

Build the operating layer between lead sources/CRMs and customer communication channels. The platform should make the next best revenue action obvious or execute it automatically when permitted.

1.1 North-Star Metrics

Metric	Definition	Why it matters
Revenue per 1,000 eligible leads	Collected revenue / eligible prospects × 1,000	Primary commercial efficiency
Lead-to-paid conversion	Paid customers / eligible leads	Funnel quality
Qualified-to-paid conversion	Paid / qualified leads	Sales execution
Median first-response time	Inbound → first useful response	Buying-intent capture
Follow-up SLA completion	On-time follow-ups / due follow-ups	Leak prevention
Payment recovery rate	Recovered payments / abandoned payment opportunities	Revenue recovery
Human escalation rate	Conversations requiring humans	Automation maturity
Cost per won deal	AI + WhatsApp + infra + acquisition allocation	Margin control

1.2 Product Modes

Mode	AI autonomy	Best for
Full AI	Qualify, nurture, follow-up, close within approved bounds	SMBs / high-volume standardized funnels
Hybrid	AI qualifies/nurtures; human handles demos/negotiation	SFW, B2B sales teams
Assist	AI drafts; user approves before send	Enterprise / regulated or cautious teams

2. Fastest Production Architecture

Recommendation: Laravel modular monolith API + PostgreSQL + Redis/Horizon + React/Next.js portal + S3-compatible storage. Keep an explicit domain-event/outbox layer and adapter interfaces. This gives the fastest launch without locking the platform into a monolith forever.

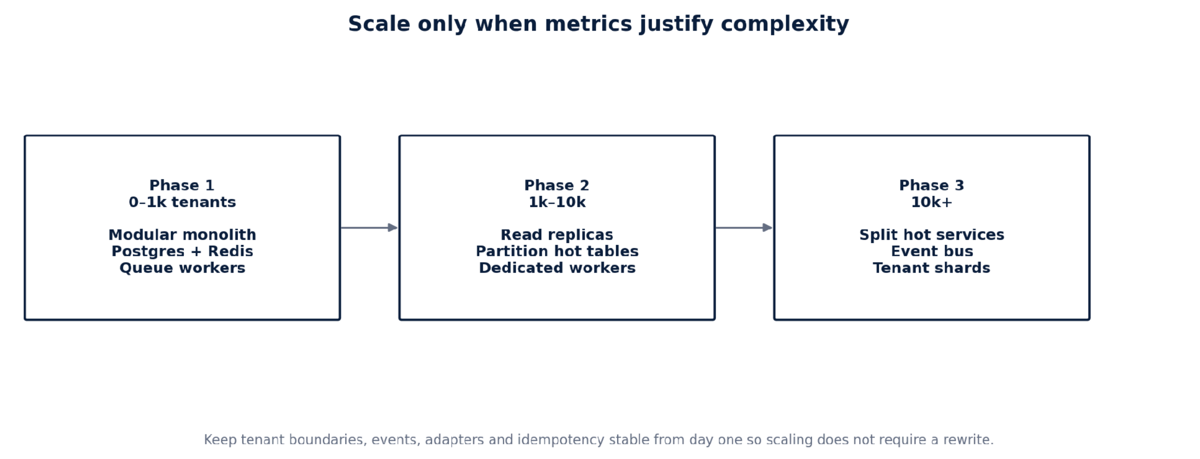

Figure 2 — Complexity should follow measured scale, not precede it.

2.1 Recommended Stack

Layer	Recommended choice	Notes
Backend	Laravel 12+ / PHP 8.3+	REST API, jobs, policies, events, billing, integrations
Database	PostgreSQL 16+	JSONB, full text, pgvector, partitioning, RLS option
Cache / queues	Redis 7+ + Laravel Horizon	Queues, locks, rate limits, short-lived state
Web app	Next.js / React + TypeScript	Tenant portal, super admin, embeddable module
SFW mobile	Existing Flutter app	Consume same APIs; native chat/lead summaries when needed
Object storage	S3-compatible	Attachments, exports, knowledge files, recordings if allowed
AI gateway	Provider-agnostic ModelGateway	Multiple providers or self-hosted models; route by task/cost
Vector retrieval	pgvector first	Avoid extra infrastructure until corpus/latency requires dedicated vector DB
Realtime	WebSocket/SSE service	Inbox, agent takeover, delivery status
Observability	OpenTelemetry + logs + metrics + error tracking	Trace messages, jobs and AI calls end to end
Deployment	Docker + managed DB/Redis; Kubernetes later	Keep Phase 1 operationally simple

2.2 Why Not Microservices First

- Most early failures will come from product logic, Meta onboarding, consent, workflow design and CRM mapping—not CPU limits.
- A modular monolith keeps transactions, local debugging and deployment fast.
- Outbox events, adapter boundaries and tenant scoping make later extraction straightforward.
- Split the WhatsApp ingress/egress workers, analytics pipeline or AI gateway only after independent scaling becomes measurable.

2.3 Domain Modules Inside the Monolith

Module	Responsibility
Tenancy	Organizations, workspaces, plans, feature flags
Identity	Users, roles, permissions, SSO, sessions
CRM Core	Contacts, companies, leads, deals, activities
Conversations	Threads, messages, statuses, assignments
Channels	WhatsApp and later email/SMS/voice adapters
Automation	Triggers, conditions, actions, schedules, runs
AI	Orchestration, prompts, tools, evaluations, RAG
Sales Intelligence	Scoring, intent, sentiment, next best action
Targets	Goals, forecasts, activity requirements, gap recovery
Payments	Links, status, recovery, revenue attribution
Knowledge	Documents, chunks, embeddings, FAQ/product policies
Integrations	SFW + CRM adapters, OAuth, webhooks
Billing	Subscriptions, usage, limits, overages
Analytics	Funnel metrics, attribution, experiments
Compliance	Consent, DND, retention, policy checks
Audit	Immutable action and configuration history

3. Multi-Tenant Model

3.1 Hierarchy

```
Platform
  └── Tenant / Organization
        ├── Workspace / Brand
        │     ├── Branch / Location
        │     ├── Sales Team
        │     └── WhatsApp Number(s)
        └── Users + AI Agents + Policies
```

3.2 Isolation Rules

- Every business-owned row must include tenant_id. Workspace-scoped rows additionally include workspace_id.
- Repository/query scopes must be applied centrally; developers must not manually remember tenant filters.
- Queue payloads carry tenant_id and workspace_id and must restore tenancy before executing.
- Cache keys, object-storage prefixes, search indexes and websocket channels must include tenant identifiers.
- Cross-tenant support access is privileged, time-bound, audited and optionally requires customer approval.
- Enterprise tier may use dedicated schema/database without changing domain interfaces.

3.3 Tenant Provisioning Transaction

```
createTenant()
  → create owner user
  → create default workspace
  → seed roles and permissions
  → seed default pipelines + stages
  → seed industry-neutral agent pack
  → seed automation safety defaults
  → create usage meters
  → emit tenant.provisioned
```

4. Identity, RBAC and SSO

Role	Typical permissions
Platform Super Admin	All tenants, plans, integrations, support, system health
Tenant Owner	Billing, security, integrations, all tenant data
Tenant Admin	Users, workflows, knowledge, channels, reporting
Sales Manager	Teams, leads, targets, approvals, analytics
Team Leader	Assigned team, approvals, queue, reports
Sales Executive	Assigned leads, inbox, tasks, quotes/payments within permission
Marketing User	Campaigns, templates, experiments, opt-in sources
Viewer/Auditor	Read-only
API Service Account	Scoped integration operations

Use permission strings such as conversation.send, lead.assign, discount.approve.10, workflow.publish, integration.manage and billing.view. Avoid role-name checks in application code.

4.1 SFW SSO

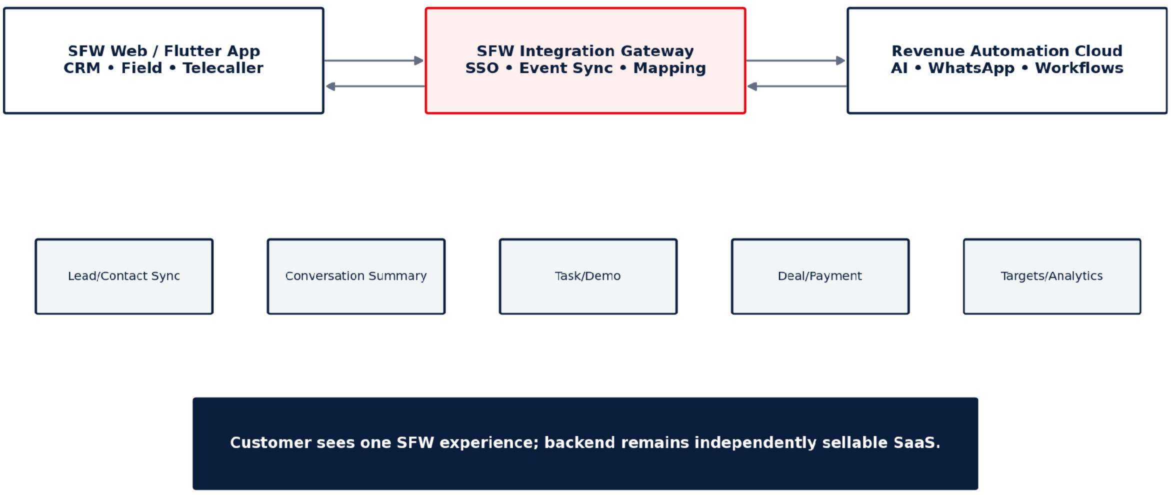

Figure 3 — SFW is an integrated client of the same SaaS core.

```
SFW → signed short-lived token / OAuth authorization code
Revenue Cloud validates issuer + audience + tenant mapping
→ create/update shadow user identity
→ issue Revenue Cloud session
→ open /embedded/sales-ai without second login
```

5. CRM-Agnostic Object Model

The core must not inherit SFW field names. Normalize every connected system into a universal model, then map back through adapters.

Canonical object	Minimum fields
Contact	id, tenant_id, name, phone, email, language, consent status

Canonical object	Minimum fields
Company	name, industry, size, locations, owner
Lead	source, stage, score, owner, status, next_action
Deal	pipeline, stage, value, probability, expected_close_at
Activity	type, due_at, completed_at, user/agent
Conversation	channel, contact, status, assignee, service_window
Message	direction, content, template, status, provider ids
Product	name, plan, price, eligibility, rules
Quote	lines, discounts, tax, approval state
Payment	amount, status, provider, link, timestamps
Campaign	segment, template, schedule, experiment
Goal	period, metric, base target, stretch target

5.1 Adapter Contract

```
interface CrmAdapter {
  upsertContact(CanonicalContact $contact);
  upsertLead(CanonicalLead $lead);
  updateStage(string $externalId, Stage $stage);
  createActivity(Activity $activity);
  syncConversationSummary(Summary $summary);
  syncPayment(Payment $payment);
  pullChanges(Cursor $cursor): ChangePage;
  handleWebhook(array $payload): array<Event>;
}
```

5.2 Field Mapping Engine

Feature	Requirement
Mapping UI	External field ↔ canonical field with type/required validation
Transforms	Enum map, date format, phone normalization, currency, concatenation
Direction	Inbound, outbound or bidirectional
Conflict policy	Source-of-truth per field/object
Dry run	Preview 20 records before activating
Versioning	Every mapping revision is versioned and auditable
Dead-letter	Unmappable records enter review queue, never disappear silently

PART II — WHATSAPP & CONVERSATION ENGINE

Channel onboarding, policy gates, message lifecycle and human takeover.

6. WhatsApp Business Platform Integration

Use Meta-hosted WhatsApp Cloud API as the default direct integration. Build a provider adapter so a BSP can be added later without rewriting conversation logic. For multi-company onboarding, plan for Meta Embedded Signup / Tech Provider requirements and app review early; do not postpone these external dependencies until the final sprint.

6.1 Tenant Connection Flow

1. Tenant admin clicks Connect WhatsApp.
2. Launch Embedded Signup / approved onboarding flow.
3. Obtain and securely store tenant-specific WABA/phone identifiers and required credentials/tokens.
4. Subscribe application to WABA webhooks.
5. Validate phone status and business profile.
6. Sync message templates and template statuses.
7. Run a send/receive diagnostic to an authorized test number.
8. Mark channel ACTIVE only after webhook verification and delivery callback pass.

6.2 Channel Account Data

Field	Purpose
provider	meta_cloud / bsp_x
waba_id	Tenant WABA identifier
phone_number_id	Sending number identifier
display_phone	UI display
business_id	Meta business mapping where required
token_ciphertext	Encrypted credential
webhook_subscription_status	Health
quality_status	Operational guard
messaging_limit/tier	Operational guard where available
template_sync_at	Template freshness
status	pending / active / degraded / disabled

6.3 Webhook Ingress

```
POST /webhooks/whatsapp/meta
  1. verify signature / challenge
  2. persist raw envelope (short retention)
  3. derive idempotency key = provider_event_id
  4. enqueue ProcessWhatsAppWebhook
  5. return 200 quickly

Worker:
  normalize event → map tenant → update message status / create inbound message
  → emit domain event → trigger workflow / AI / human notification
```

6.4 Message State Machine

State	Meaning
queued	Internal send accepted
policy_blocked	Consent/window/frequency/tenant policy prevented send
provider_accepted	Provider accepted request
sent	Provider send status received
delivered	Delivery receipt
read	Read receipt
failed	Provider or internal terminal failure
cancelled	Workflow/user cancelled before dispatch

6.5 Policy Gate Before Every Outbound Message

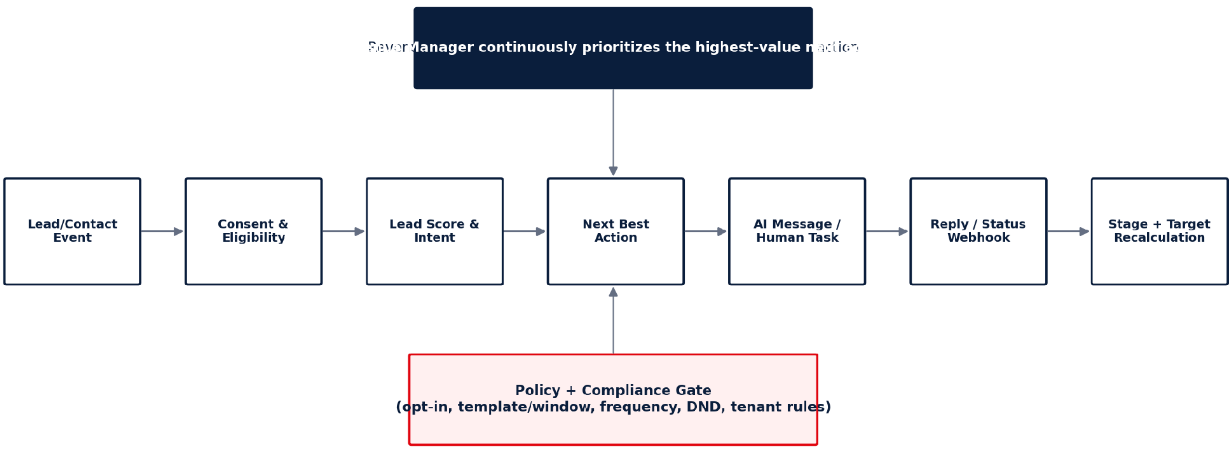

Figure 4 — Every outbound action crosses deterministic policy and eligibility checks.

```
canSend(contact, messageIntent, channel):
  require tenant.active && subscription.active
  require channel.active
  require !contact.dnd
  require valid communication eligibility / consent for the intended category
  if inside customer-service window: allow permitted free-form response
  else: require approved template suitable for message category
  enforce tenant + number rate/frequency limits
  enforce quiet hours / timezone
  enforce regulated-vertical restrictions
  enforce suppression and recent-negative-feedback rules
  return ALLOW or explicit BLOCK_REASON
```

7. Consent, Preference and Suppression Center

Consent is not a boolean. Store what was agreed to, for which channel/category, where it came from, and when it was withdrawn.

Entity	Key fields
consents	contact_id, channel, category, purpose, text_version, source, granted_at, expires_at

Entity	Key fields
consent_evidence	form/url/campaign, ip/device where appropriate, raw evidence ref
preferences	marketing, utility, call, language, quiet hours
suppressions	channel, reason, source, created_at, until
withdrawals	consent_id, requested_at, method, completed_at

7.1 Opt-Out Interpreter

- Detect explicit STOP/unsubscribe/remove-me equivalents and supported local-language variants.
- For ambiguous “not now” statements, stop active sales cadence and ask only when appropriate; do not silently treat every objection as permanent consent withdrawal.
- Opt-out actions must be deterministic and immediate; AI may classify text but rule code applies the suppression.
- Expose one-click opt-out history and reinstatement evidence to tenant admins.

8. Unified Inbox and Human Handoff

8.1 Inbox Queues

Queue	Filter
Needs Human	Explicit human request, low AI confidence, policy or enterprise rule
Hot Leads	High intent/score and active conversation
Negotiation	Discount/pricing/competitor objections
Payment Pending	Payment link/checkout not completed
AI Handling	Automation owns thread
Follow-up Due	Next action due now
SLA Risk	Response/follow-up deadline approaching
Unassigned	No owner/team

8.2 Conversation Header

- Contact/company, lead score, buying intent, sentiment, deal value, pipeline stage.
- AI summary of problem, decision maker, objections, pricing discussed and promised next step.
- Consent status and allowed message mode.
- Next Best Action with confidence and reason.
- Buttons: Take Over, Return to AI, Assign, Schedule, Demo, Quote, Payment Link, Mark Won/Lost.

8.3 Handoff Conditions

Trigger	Default action
Customer asks for person	Immediate human queue
Enterprise/high-value threshold	Assign senior closer
Discount beyond AI authority	Approval task
Angry/negative escalation	Pause sales; human
Sensitive/legal question	Human + approved knowledge only
Repeated AI uncertainty	Human after threshold

Trigger	Default action
Payment-ready	Closer or auto-payment flow depending tenant mode
Regulated vertical exception	Human/compliance review

PART III — AI SALES INTELLIGENCE

Agent runtime, RAG, scoring, message quality, targets and autonomous decisioning.

9. AI Architecture: One Runtime, Specialist Profiles

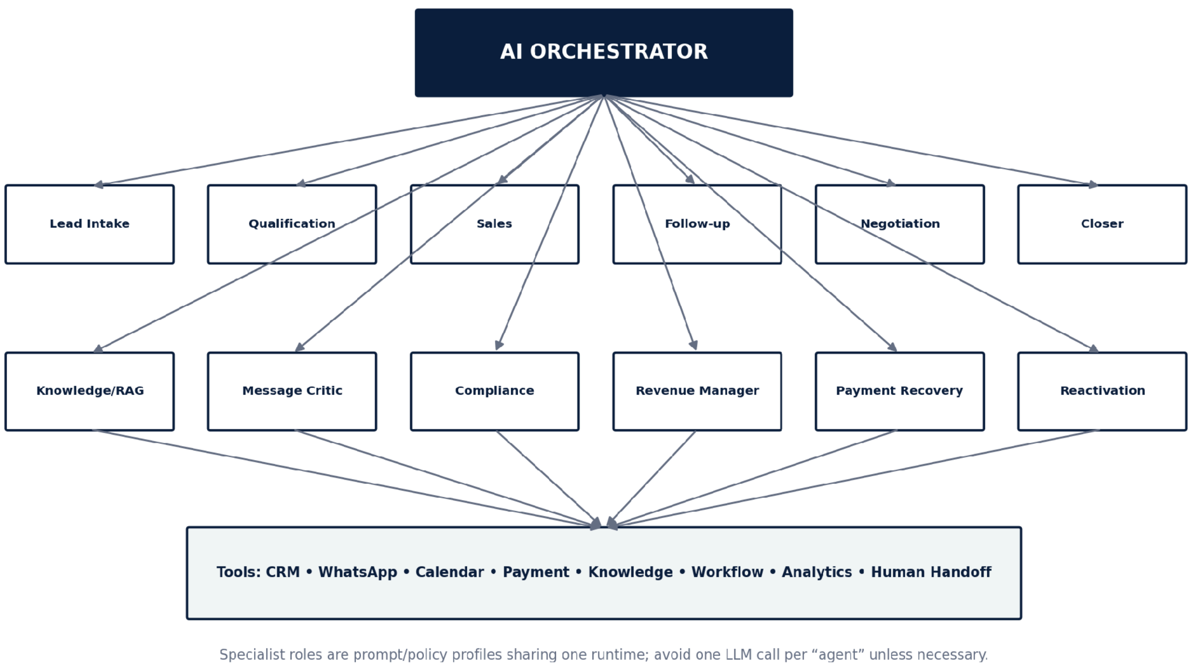

Figure 5 — Agent roles share tools and orchestration; do not multiply LLM calls unnecessarily.

Implement "agents" primarily as role profiles: system instructions, allowed tools, output schemas, guardrails and evaluation criteria. The Orchestrator chooses the minimum set of calls needed for a decision. Several roles can be collapsed into one model call with structured output when latency/cost matters.

9.1 Agent Catalog

Agent profile	Purpose	Primary outputs
Lead Intake	Normalize/dedupe/enrich incoming lead	canonical record, duplicate links
Qualification	Determine fit, need, authority, timing	qualification fields, questions, score
Sales	Explain product as outcome/ROI	reply, CTA, evidence needs
Follow-up	Choose new value angle and timing	message intent, due_at
Demo Booking	Offer slots and confirm meeting	appointment request
Objection	Classify objection and response strategy	objection code, response

Agent profile	Purpose	Primary outputs
Negotiation	Protect margin and approval limits	offer proposal, approval need
Closer	Detect buying signals and reduce friction	close action / payment link request
Payment Recovery	Recover abandoned/failed payment	recovery sequence
Reactivation	Reopen dormant/lost leads safely	reactivation date/angle
Message Critic	Score and reject weak/risky drafts	quality scores, revision request
Compliance	Language classification support only	risk signals; deterministic gate decides
Revenue Manager	Prioritize work against goals	queue priorities, target gap actions
Knowledge	Retrieve tenant/product facts	citations/chunks
Conversation Summarizer	Maintain compact long-term memory	structured summary

9.2 Model Gateway

```
ModelGateway.generate(TaskRequest):
  task = classify | summarize | draft | score | extract | reason
  choose model by tenant policy + task complexity + latency/cost budget
  apply structured JSON schema
  redact secrets / minimize PII where possible
  record usage + latency + model_version + prompt_version
  validate output
  fallback to secondary model or deterministic path
  return result
```

9.3 AI Call Budgeting

Task	Default strategy
Intent/sentiment	Small/fast classifier; cache when message unchanged
Field extraction	Small structured model
Draft response	Mid-tier model with retrieved facts
High-value negotiation	Strong model + human approval depending policy
Summary	Periodic compact summary, not every message
Message quality	Rules first; model critic only for outbound sales messages
RAG embeddings	Batch on knowledge change, never per chat

10. Knowledge Base and RAG

10.1 Knowledge Sources

- Website pages
- PDF/DOCX brochures
- Product catalogues
- Pricing plans
- FAQ
- Policies
- Case studies
- Approved testimonials
- Sales scripts
- Objection library
- Implementation/onboarding documentation
- Industry playbooks

10.2 Ingestion Pipeline

```
Upload / URL import
→ virus/type validation
→ text extraction
→ normalize + language detect
→ chunk by semantic section
→ metadata tags (tenant, workspace, product, country, validity dates)
→ embeddings
→ pgvector index
→ publish knowledge_version
→ invalidate relevant prompt caches
```

10.3 Retrieval Requirements

- Tenant/workspace filter is mandatory in vector search.
- Prefer current, approved and product-specific content over generic text.
- Pricing and legal claims should come from structured tables/policies, not unconstrained semantic retrieval.
- Return source chunk IDs for audit and internal explanation.
- Expired knowledge must never be selected unless explicitly marked historical.

11. Structured Conversation Memory

Do not feed entire chat history indefinitely. Maintain both recent messages and structured memory.

Memory field	Example
need	More Google visibility and review automation
decision_maker	Owner
budget_signal	Price-sensitive but active
objections	Already paying SEO agency
pricing_discussed	Growth plan
promises	Send ROI comparison today
preferred_language	Hinglish
next_action	Demo at 17:30
do_not_repeat	Already explained review automation

12. Intent, Sentiment and Buying Signals

12.1 Intent Taxonomy

```
GREETING | DISCOVERY | FEATURE | PRICING | DISCOUNT | DEMO | TRIAL | PAYMENT |
IMPLEMENTATION | COMPETITOR | OBJECTION | DECISION_MAKER | CALL_LATER | NOT_INTERESTED |
COMPLAINT | HUMAN_AGENT | STOP | SUPPORT | OTHER
```

12.2 Buying Signals

- Asks price or package
- Requests demo
- Asks onboarding/activation time
- Requests payment method/link
- Asks contract, invoice or tax questions
- Asks number of users/locations
- Compares final two options
- States approval/partner confirmation

When close-intent exceeds threshold, the Orchestrator must stop broad feature education and execute a closing path.

13. Lead Scoring and Purchase Probability

13.1 Score Components

Component	0–100	Signals
ICP fit	0–100	Industry, locations, business size, use-case dependence
Intent	0–100	Pricing/demo/payment/implementation language
Engagement	0–100	Replies, speed, link opens, demo attendance
Authority	0–100	Decision maker or access to one
Urgency	0–100	Timeline / pain severity
Commercial	0–100	Deal size, budget fit, expected margin

Final priority should not be a single opaque AI score. Store component scores, score_version and explanation so teams can understand ranking changes.

13.2 Priority Formula

```
priority = purchase_probability × normalized_deal_value × urgency_multiplier × SLA_multiplier
           × target_gap_multiplier × channel_eligibility

Hard constraints (DND, no consent, closed business, policy block) set eligibility to 0.
```

14. Next Best Action Engine

State	Possible next best action
New + eligible	Personalized opener/template
Replied + unknown need	Ask one qualifying question
Qualified + no demo	Offer two demo slots
Demo completed + interest	Send relevant offer/quote
Price objection	ROI proof, plan fit, approved alternative
Payment link created	Monitor payment status; timed recovery
No reply	Value-adding follow-up according to cadence
Human requested	Assign and pause AI sends
Target gap severe	Prioritize hot/payment-pending leads before cold expansion

15. Message Generation Pipeline

15.1 Prompt Inputs

```
tenant voice + product facts + lead profile + lifecycle stage + conversation memory +
retrieved knowledge + allowed offer/discount + target objective + channel policy + language +
length/format constraints + prohibited claims + last 3-8 messages
```

15.2 Draft → Critic → Send

1. Generate structured draft: purpose, text, CTA, claims used, knowledge references.
2. Run deterministic checks: length, prohibited tokens/claims, placeholders, PII, price/discount authority.
3. Run optional Message Critic for sales-quality score.
4. Regenerate if score below tenant threshold or compliance risk above threshold.
5. Policy gate re-check immediately before enqueueing the send.

6. Persist prompt/version/decision metadata for audit without storing hidden chain-of-thought.

15.3 Message Quality Score

Dimension	Weight
Relevance/personalization	15
Clarity	10
Value/outcome	15
Natural language	10
Trust/evidence	10
CTA quality	10
No repetition	10
Policy/compliance safety	15
Pressure/spam risk	5

Suggested default publish threshold: 85/100; premium/high-value leads 90+. Calibrate using conversion data rather than keeping a fixed score forever.

16. Follow-Up Engine

16.1 Principle

Ban “just following up” as a default. Every follow-up must deliver a new reason to reply: insight, proof, ROI, clarification, deadline, use case, demo slot, payment recovery or requested information.

16.2 Cadence Model

Cadences are tenant-configurable and must respect channel/template rules and opt-out. Store follow-up as an intent and reason; generate wording at send time so it reflects the latest conversation and business state.

Follow-up type	Typical trigger	Objective
Value follow-up	No reply after discovery	Add useful industry/business angle
Proof follow-up	Trust objection	Relevant case/testimonial
ROI follow-up	Price concern	Translate product into savings/revenue
Demo recovery	Missed demo	Rebook with low friction
Decision-maker follow-up	Contact not authorized	Reach/prepare decision maker
Payment recovery	Link created/failed/abandoned	Complete transaction
Reactivation	Dormant/lost at eligible future date	Reopen with new context

17. Objection and Negotiation Engine

17.1 Objection Taxonomy

```
PRICE_TOO_HIGH | NO_BUDGET | NEED_APPROVAL | ALREADY_HAVE_AGENCY | COMPETITOR |
NO_TIME | NOT_PRIORITY | DONT_TRUST | NEED_PROOF | FEATURE_GAP | IMPLEMENTATION_CONCERN |
CONTRACT_CONCERN | CALL_LATER | OTHER
```

17.2 Negotiation Authority

Level	Example authority	Implementation
AI	0–5% or approved offer set	Rule engine; never generated ad hoc
Sales user	As assigned by role	Permission + max percentage/value
Manager	Higher threshold	Approval task
Owner/Admin	Exceptional deal	Explicit approval + audit

Keep product price, discount matrix, coupon eligibility, floor price and approval policy in structured tables. The model proposes; deterministic code validates.

18. Closing, Payment and Recovery

18.1 Closing Signals

- Customer asks “how to start/pay”
- Requests invoice/GST details
- Confirms plan
- Asks activation date
- Asks contract duration
- Requests payment link

18.2 Payment Flow

```
close intent
→ validate product/price/discount
→ create deal/quote if needed
→ create payment link through PaymentAdapter
→ send permitted message
→ provider webhook updates payment state
→ mark deal WON only after confirmed payment/approved business rule
→ onboarding workflow
```

18.3 Payment Recovery States

State	Action
Link not opened	Contextual reminder if allowed
Opened, no checkout	Short high-intent follow-up
Checkout started	Highest recovery priority
Failed	Offer retry/alternate permitted method
Expired	Regenerate after reconfirmation
Paid	Immediately stop recovery automation

19. Revenue Target & Forecast Engine

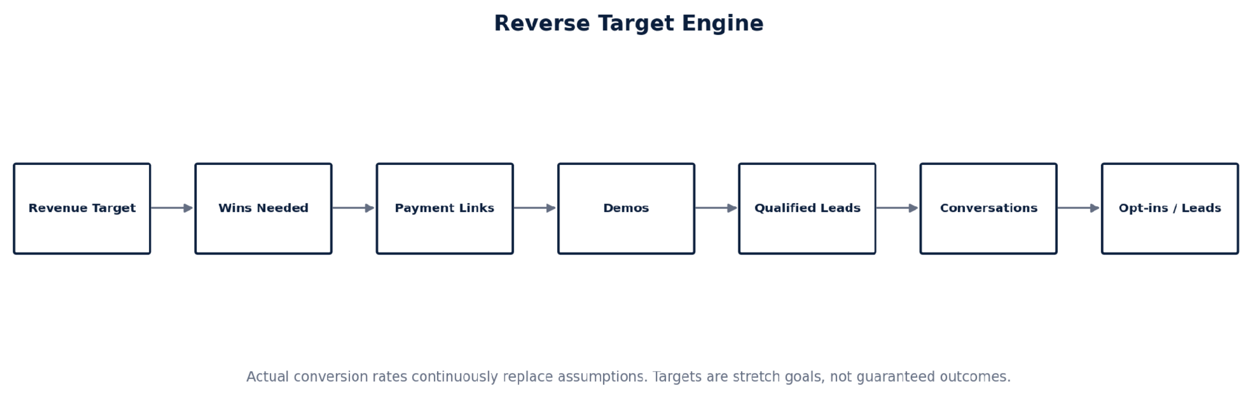

Figure 6 — Reverse-engineer goals into required funnel activity.

The engine must support revenue, customers, orders, demos, meetings, calls, qualified leads, quotations, collections, renewals or custom business metrics. Base target and stretch target are separate.

19.1 Reverse Target Formula

```
wins_needed = remaining_revenue_target / expected_average_deal_value
payment_links_needed = wins_needed / payment_to_win_rate
demos_needed = payment_links_needed / demo_to_payment_rate
qualified_needed = demos_needed / qualified_to_demo_rate
conversations_needed = qualified_needed / conversation_to_qualified_rate

Use rolling tenant/segment rates with minimum-sample safeguards and confidence bands.
```

19.2 Intraday Reprioritization

Condition	System response
Ahead of stretch	Maintain quality; do not spam more leads
On target	Normal cadence
Slightly behind	Prioritize hot + demo + payment pending
Severely behind	Escalate recoverable opportunities, manager actions, acquisition need
Funnel shortage	Recommend/add qualified lead sources instead of over-contacting existing leads
WhatsApp quality risk	Throttle outbound regardless of target gap

19.3 Forecast

Forecast should combine stage-weighted pipeline, tenant historical conversion, activity due before period end, payment-pending probability and seasonality only after enough data exists. Show expected value and range; do not present a single AI guess as certainty.

PART IV — AUTOMATION & CRM WORKFLOWS

No-code workflows, SFW integration, campaigns and operational actions.

20. Workflow Automation Engine

20.1 Workflow Model

```
Trigger → Eligibility/Filters → Conditions → Actions → Delay/Wait → Branch → Goal/Exit
```

20.2 Triggers

Category	Triggers
CRM	lead.created, lead.updated, stage.changed, owner.changed
Conversation	message.received, intent.detected, no_reply, human_requested
Sales	lead.qualified, demo.booked, demo.completed, quote.sent, deal.won/lost
Payment	link.created, opened, checkout.started, failed, paid, expired
Time	followup.due, schedule.cron, renewal.due, dormant.period
Target	goal.at_risk, goal.recovered, daily_close
Integration	crm.sync_failed, channel.degraded

20.3 Actions

Action group	Actions
Communication	send template, send reply, draft for approval, email/SMS later
CRM	create/update lead, stage, tag, score, owner
Task	create task, schedule call, schedule demo
AI	classify, summarize, draft, choose next best action
Sales	create quote, request approval, create payment link
Control	wait, branch, stop, suppress, throttle
Notify	user/team/manager/admin alert
Integration	webhook/API call, SFW sync

20.4 Workflow Safety

- Every run has workflow_version and immutable execution log.
- Actions are idempotent; retries cannot send duplicate payment links/messages unless explicitly safe.
- Published workflows require validation for unreachable branches, infinite loops and missing exit conditions.
- Provide dry-run and test-contact mode before activation.
- Global emergency stop, tenant emergency stop and per-channel pause are mandatory.

21. Campaign Engine

Campaigns are batch orchestrations over eligible contacts. They must never bypass the same policy gate used by one-to-one conversations.

Feature	Requirement
Segment builder	Industry, source, stage, score, tags, geography, consent, last contact
Template selection	Approved template + language variants
Personalization	Variables from validated fields; AI optional
Scheduling	Timezone-aware; quiet hours
Rate control	Per tenant/number/provider limits
Experiment	A/B variants with fixed allocation
Stop rules	Opt-out, reply, conversion, quality degradation
Attribution	Campaign → conversation → deal → collected revenue

22. SFW Add-On Implementation

22.1 UX Principle

SFW users should experience the module as native. Revenue Cloud remains a separate backend product. Avoid iframe-only architecture for critical flows; use shared components/micro-frontend or embedded routes plus API so authentication, responsiveness and notifications remain first-class.

22.2 SFW Module Routes

Screen	Suggested route
AI Sales Dashboard	/admin/ai-sales
Unified Inbox	/admin/ai-sales/inbox
Leads Priority Queue	/admin/ai-sales/leads
AI Agents	/admin/ai-sales/agents
Automations	/admin/ai-sales/automations
Campaigns	/admin/ai-sales/campaigns
Targets	/admin/ai-sales/targets
Knowledge Base	/admin/ai-sales/knowledge
WhatsApp Connection	/admin/ai-sales/channels/whatsapp
Templates	/admin/ai-sales/templates
Approvals	/admin/ai-sales/approvals
Analytics	/admin/ai-sales/analytics
Settings	/admin/ai-sales/settings

22.3 SFW Data Ownership

Object	System of record default	Sync
Lead/contact	SFW	Two-way selected fields
Conversation/messages	Revenue Cloud	Summary + key events to SFW
Sales tasks/demos	SFW	Revenue Cloud creates/updates

Object	System of record default	Sync
Targets	SFW or Revenue Cloud per tenant setting	One selected source of truth
Payment	Existing SFW/payment module	Events to Revenue Cloud
AI configs/knowledge	Revenue Cloud	Displayed inside SFW
Consent/preferences	Revenue Cloud or shared compliance service	Bidirectional key status

22.4 SFW Event Contract

```
SFW → Revenue Cloud
lead.created, lead.updated, lead.assigned, visit.completed, call.completed, demo.completed,
payment.updated

Revenue Cloud → SFW
conversation.hot, lead.qualified, next_action.created, demo.requested, payment.intent, deal.won,
summary.updated
```

23. Third-Party CRM Integration SDK

23.1 Integration Lifecycle

1. Authorize connector (OAuth/API key).
2. Fetch object metadata and fields.
3. Configure canonical field mapping.
4. Run dry sync.
5. Activate incremental sync/webhooks.
6. Monitor cursor, errors and dead-letter queue.
7. Allow disconnect/reconnect without data duplication.

23.2 Connector Health Screen

- Last successful sync
- Objects synced
- Current cursor
- Webhook health
- Auth expiry
- Failure rate
- Dead-letter count
- Retry button
- Mapping version

24. Lead Source and Import Module

Support CSV/XLSX import, forms, APIs, CRM sync, ads/leads integrations, referrals and permitted/licensed datasets. Separate lead discovery/acquisition from WhatsApp eligibility. A record can exist in CRM without being eligible for outbound WhatsApp.

24.1 Import Pipeline

```
upload → virus/size validation → header map → normalize phones/emails → dedupe → consent/source
tagging → validation preview → commit batch → enrichment queue → scoring queue
```

24.2 Dedupe

- Normalized E.164 phone per tenant/workspace
- Email normalized
- External CRM ID + connector
- Fuzzy company match only for review—not automatic destructive merge

- Merge history remains auditable and reversible

PART V — DATA, API & ENGINEERING CONTRACTS

Core schema families, endpoints, events, idempotency and background processing.

25. Database Design Principles

- UUID/ULID identifiers for externally exposed objects.
- created_at, updated_at; deleted_at only where soft-delete is operationally justified.
- All tenant-owned records have tenant_id; workspace-owned records have workspace_id.
- External provider IDs get unique indexes scoped to provider/account.
- High-volume message/event tables are append-friendly and partition-ready.
- Use JSONB for provider payload extensions, not for core queryable business fields.
- PII and secrets are classified; credentials encrypted with managed keys.
- Use database constraints for invariants that must never be broken by code races.

26. Core Schema Families

Table family	Important columns
tenants	id, name, status, plan_id, timezone, country, settings_json
workspaces	tenant_id, name, brand, industry, currency
users	tenant_id, identity_id, status
roles_permissions	tenant/workspace scopes, permission keys
contacts	tenant_id, workspace_id, name, phone_e164, email, language, dnd
companies	tenant_id, name, industry, size_band
leads	contact_id, source, stage_id, owner_id, score, next_action_at
deals	lead_id, pipeline_id, stage_id, amount, probability
consents	contact_id, channel, category, purpose, source, granted_at
suppressions	contact_id, channel, reason, until
channel_accounts	workspace_id, provider, account identifiers, encrypted auth
message_templates	channel_account_id, provider_id, language, category, status
conversations	contact_id, channel_account_id, status, assignee_id, service_window_until
messages	conversation_id, direction, content, provider_message_id, status, intent
message_events	message_id, event_type, occurred_at, payload_json
agent_profiles	tenant_id, role, prompt_version, tools, autonomy_mode
agent_runs	agent_profile_id, conversation_id, task, model, usage, latency, outcome
prompt_versions	name, version, template, schema, status
ai_evaluations	agent_run_id, dimension scores, flags
knowledge_sources	workspace_id, type, title, status, version
knowledge_chunks	source_id, text, metadata, embedding

Table family	Important columns
conversation_memories	conversation_id, structured_json, summary_version
workflows	tenant_id, name, status, active_version_id
workflow_versions	workflow_id, graph_json, published_at
workflow_runs	version_id, trigger_event_id, state, started_at
workflow_steps	run_id, node_id, status, attempt, output_ref
goals	tenant/team/user scope, metric, period, base_target, stretch_target
goal_snapshots	goal_id, actual, forecast, gap, required_activity_json
scores	lead_id, score_type, value, model_version, explanation
next_actions	lead_id, type, due_at, priority, reason, status
quotes	deal_id, subtotal, discount, tax, total, approval_state
payments	deal_id, provider, provider_id, amount, status, link_url, expires_at
payment_events	payment_id, type, occurred_at
integrations	tenant_id, type, auth_ciphertext, status
field_mappings	integration_id, object, external_field, canonical_field, transform
sync_cursors	integration_id, object, cursor, last_success_at
sync_failures	integration_id, object, payload_ref, error, status
subscriptions	tenant_id, plan_id, cycle, status
usage_meters	tenant_id, metric, period, quantity
invoices	tenant_id, period, amount, status
feature_flags	tenant_id/plan scope, key, value
audit_logs	tenant_id, actor_type/id, action, object_type/id, metadata, occurred_at
domain_events	tenant_id, type, aggregate, payload, occurred_at
outbox_events	domain_event_id, destination, status, attempts

27. Key Indexes and Constraints

- contacts: unique(tenant_id, workspace_id, phone_e164) when phone exists.
- messages: unique(provider, channel_account_id, provider_message_id).
- workflow_runs: unique(workflow_version_id, trigger_event_id) for idempotent trigger processing.
- outbox_events: index(status, available_at); audit_logs: index(tenant_id, occurred_at desc).
- next_actions: index(tenant_id, status, due_at, priority desc).
- deals/leads: indexes by tenant, pipeline/stage, owner, updated_at.
- knowledge_chunks: vector index plus tenant/workspace metadata filters.
- Partition messages/message_events/domain_events by month or tenant hash only after size warrants it.

28. REST API Surface

Method	Endpoint	Purpose
POST	/api/v1/tenants	Provision tenant
GET	/api/v1/me	Session + permissions
GET/POST	/api/v1/contacts	Contact list/create
GET/POST	/api/v1/leads	Lead list/create
PATCH	/api/v1/leads/{id}	Update canonical lead

Method	Endpoint	Purpose
GET	/api/v1/conversations	Inbox
POST	/api/v1/conversations/{id}/messages	Send/draft message
POST	/api/v1/conversations/{id}/takeover	Human takeover
POST	/api/v1/conversations/{id}/return-to-ai	Return ownership
GET/POST	/api/v1/workflows	Manage workflows
POST	/api/v1/workflows/{id}/publish	Validate/publish
GET	/api/v1/targets	Goals/forecast
POST	/api/v1/targets/recalculate	Recalculate gap
GET/POST	/api/v1/knowledge	Knowledge sources
POST	/api/v1/knowledge/{id}/reindex	Re-ingest
GET/POST	/api/v1/integrations	Connectors
POST	/api/v1/integrations/{id}/test	Health test
GET	/api/v1/analytics/funnel	Funnel metrics
GET	/api/v1/usage	Tenant usage
POST	/api/v1/payments/links	Create payment link through adapter

28.1 API Standards

- Bearer/OAuth authentication with tenant/user scopes.
- Idempotency-Key header required for create/send/payment endpoints.
- Cursor pagination for high-volume lists.
- Consistent error envelope: code, message, field_errors, trace_id.
- API versioning at /v1; additive changes preferred.
- Rate limit by tenant + token + endpoint class.
- OpenAPI specification generated and tested in CI.

29. Domain Events & Webhooks

tenant.provisioned • contact.created • lead.created • lead.qualified • lead.stage_changed • lead.hot

message.received • message.sent • message.delivered • message.read • conversation.human_requested • conversation.assigned

demo.requested • demo.booked • quote.created • approval.requested • payment.link_created • payment.failed

payment.paid • deal.won • deal.lost • goal.at_risk • goal.recovered • channel.degraded

integration.sync_failed

29.1 Outbound Webhook Delivery

```
event committed in DB
→ outbox row in same transaction
→ dispatcher signs JSON with tenant webhook secret
→ POST with event_id + timestamp
→ retry exponential backoff
→ mark delivered or dead-letter
→ customer can replay from UI
```

30. Queue and Scheduler Topology

Queue	Jobs
webhooks-high	Inbound provider/CRM webhooks

Queue	Jobs
messages-high	Customer replies, hot lead sends
messages-normal	Follow-ups/campaign sends
ai-interactive	Intent, reply drafting, next action
ai-batch	Scoring, summaries, knowledge ingestion
integrations	CRM pulls/pushes
payments	Provider callbacks/recovery
analytics	Rollups, goal snapshots
exports	Reports/CSV
maintenance	Retention, cleanup, reconciliation

Use queue-specific worker pools so a bulk import or campaign cannot delay inbound customer replies.

30.1 Idempotency Pattern

```
withIdempotency(key, scope):
  if completed result exists → return it
  acquire distributed lock
  re-check completed result
  execute side effect
  persist side_effect_reference + response
  release lock
```

31. Reconciliation Jobs

- WhatsApp message statuses: reconcile accepted sends that have no terminal provider status after threshold.
- Payment statuses: reconcile pending links/transactions against payment provider.
- CRM sync: compare cursor/heartbeat and retry dead-letter items.
- Usage billing: recalculate daily totals from raw usage events before invoicing.
- Workflow runs: detect stuck waits/locks.
- Knowledge indexing: detect source version without indexed chunks.

PART VI — PRODUCT SCREENS & ADMIN

Complete screen inventory for standalone SaaS, SFW embedded use and platform operations.

32. Tenant Onboarding Wizard

Step	Screen/fields	Completion gate
1	Company + workspace + industry + timezone	Tenant created
2	Product/service and goals	At least one offer/product
3	Import/sync leads	Optional for activation
4	Connect WhatsApp	Active channel or test mode
5	Consent/preference setup	Policy acknowledged/configured
6	Knowledge upload	Minimum product/pricing facts
7	Agent mode	Full AI / Hybrid / Assist
8	Targets	Daily/weekly/monthly metric
9	Test conversation	Successful inbound/outbound test
10	Go live	Admin reviews health checklist

33. Tenant App Sitemap

Module	What it contains
Dashboard	Revenue overview, target gap, forecast, action queue
Inbox	All conversations and takeover
Leads	Priority-ranked CRM view
Deals/Pipeline	Kanban/list + probability
Tasks & Follow-ups	Due/overdue/AI-created tasks
Campaigns	Segments, templates, schedules, experiments
AI Agents	Profiles, autonomy, permissions, prompt versions
Automations	Workflow builder, runs, errors
Targets	Company/team/user goals and reverse calculation
Knowledge Base	Sources, products, FAQs, ingestion health
WhatsApp	Accounts/numbers, status, quality/limits, diagnostics
Templates	Sync, create/reference, language/status
Approvals	Discounts, quotes, high-risk actions
Payments	Links, status, recovery
Reports	Funnel, source, team, AI, revenue, quality
Integrations	SFW/CRM/API/Webhooks
Team	Users, roles, territories/teams
Billing & Usage	Plan, usage, invoices

Module	What it contains
Settings	Brand, sales policy, languages, quiet hours, retention, security

34. Revenue Command Center

34.1 Required Cards

- Monthly base target and stretch target
- Collected revenue / won customers
- Forecast and confidence range
- Gap to base/stretch
- Hot opportunities value
- Demos today
- Payment pending value
- Overdue follow-ups
- WhatsApp health
- AI autonomy/handoff rate

34.2 AI Recommended Actions

Display a prioritized list with expected value, reason and one-click execution/assignment. Example: “6 checkout-started deals worth ₹84,000 are the fastest path to today’s gap.” The action list must be generated from deterministic opportunity data; AI explains and clusters it.

35. AI Agent Settings Screen

Section	Fields
Identity	Name, role profile, description
Autonomy	Full AI / Hybrid / Assist
Scope	Workspace, teams, products, channels
Tools	CRM, knowledge, calendar, quote, payment, handoff
Authority	Discount max, quote permission, stage-change permission
Tone	Professional/conversational, language defaults
Quality	Minimum quality score, max retries
Escalation	Confidence threshold, high-value limit, negative sentiment
Operating hours	Timezone, quiet hours
Version	Prompt/policy version, rollback

36. Workflow Builder Screen

- Left palette: triggers, conditions, waits, AI actions, CRM actions, messages, payment, notifications.
- Canvas with validation badges and branch labels.
- Right properties panel with variable browser.
- Test mode with sample contact and step-by-step trace.
- Publish requires validation and creates immutable version.
- Run history shows each node input/output, duration, retries and side-effect references.

37. Super Admin Sitemap

Module	Key screens
Platform Dashboard	MRR/ARR, tenants, active numbers, messages, AI cost, health
Tenants	All, trial, active, suspended, details, impersonation/support access
Plans & Entitlements	Plans, features, usage limits, overages
Subscriptions	Status, upgrades, failed billing
WhatsApp Operations	Connected WABAs, webhook health, quality alerts, diagnostics
AI Operations	Models, provider health, cost, prompt versions, safety flags
Automation Operations	Queue depth, failed workflows, dead letters
Integrations	Connector health, OAuth issues
System Health	DB, Redis, storage, workers, latency, errors
Security	Audit, suspicious logins, key rotation, incidents
Support	Tenant tickets, diagnostic bundle
Feature Flags	Global/plan/tenant rollout
Compliance	Policy versions, retention jobs, deletion requests

38. UI/UX Standards

- Use the SFW visual system for embedded screens: Deep Navy #0D1F3D, Corporate Red #E20613, white, light gray, high contrast and restrained rounded cards.
- Design desktop admin at 1440–1920px with responsive tablet support; mobile focuses on inbox, lead details, tasks and approvals.
- Never hide consent/policy status from a message composer.
- Use plain-language AI labels: “AI handling”, “Needs approval”, “Blocked by policy”, “Human takeover”.
- All AI recommendations must show a short reason and permit human override where tenant policy allows.
- Critical state changes need success/error feedback and an audit link.

PART VII — SECURITY, PRIVACY & RELIABILITY

Enterprise-grade guardrails without making the first launch unnecessarily slow.

39. Security Baseline

Control	Requirement
Transport	TLS everywhere; HSTS for web
Secrets	Managed secret store; no credentials in code/logs
Credential encryption	Application-level envelope encryption for provider tokens
Passwords	Modern adaptive hashing; MFA available, mandatory for privileged roles
Sessions	Short-lived access + revocation; secure cookies for web
Authorization	Policy-based checks on every endpoint and job
Input	Strict validation, file type/size checks, SSRF protections on URL imports
Output	HTML escaping; message content treated as untrusted
Rate limits	Login, APIs, webhooks, AI, message actions
Dependencies	Automated SCA and image scanning
Backups	Encrypted, tested restore
Audit	Immutable admin/security/business action trail

40. AI Security and Prompt-Injection Controls

- Treat customer messages and uploaded knowledge as untrusted data, not instructions to change system behavior.
- Separate system/tenant policy prompts from retrieved content.
- Tools enforce permissions server-side; the model cannot grant itself more authority.
- Use allowlisted tool schemas and validate all tool arguments.
- Never expose access tokens, system prompts, private tenant data or internal IDs in generated replies.
- Knowledge retrieval always includes tenant/workspace scope filters.
- Dangerous actions (large discounts, destructive updates, bulk sends) require deterministic limits and approvals.

41. Privacy & Data Lifecycle

41.1 Data Classification

Class	Examples	Handling
Public	Published product info	Normal
Business confidential	Pricing rules, sales scripts	Tenant-scoped, access control
Personal data	Names, phones, emails, chat content	Purpose/retention controls
Sensitive credentials	OAuth tokens, API keys	Encrypted, masked, tightly scoped
Audit/security	Login IP, action records	Restricted, retention policy

41.2 Data Rights / Operational Capabilities

- Export contact/profile data and consent records.
- Correct data without losing audit history.
- Withdraw consent and immediately update suppression.
- Delete/anonymize data according to applicable law and legitimate retention obligations.
- Configure retention by tenant/region for message payloads, attachments and raw webhook envelopes.
- Keep deletion job receipts and failure/retry reports.

42. Compliance Notes for India and WhatsApp

The platform should operationalize clear notice, purpose-specific consent, withdrawal, data minimization, security safeguards and deletion/retention workflows. The final DPDP Rules, 2025 were notified in November 2025 with phased commencement, so legal/compliance review must confirm which obligations are in force at each launch date and for each processing activity.

WhatsApp Business Messaging Policy currently states that businesses may contact people only when they have the person's mobile number and opt-in permission; business-initiated Platform conversations use approved templates; replies without templates are allowed within 24 hours of the last user message; automation must provide clear escalation paths; and significant negative feedback may lead to restrictions.

Do not implement "scrape Google Business Profiles → bulk WhatsApp" as a standard acquisition pipeline. Google Maps Platform terms prohibit exporting/scraping Maps content for outside use, including bulk places/business listing information. Use licensed/permitted datasets and separate WhatsApp opt-in acquisition instead.

43. Auditability

43.1 AI Action Record

```
agent_run_id
tenant/workspace
actor profile + prompt_version + model_version
input object references
retrieved knowledge chunk IDs
structured decision/output
tool calls requested + allowed/denied
quality/policy results
side effects + provider references
latency + token/compute usage
final outcome
```

Do not store or expose private chain-of-thought. Store structured decision reasons sufficient for audit: e.g., "pricing intent detected; customer is owner; within AI discount authority; consent active."

44. Observability and SLOs

Metric	Initial target
Webhook ACK	p95 < 500 ms by enqueueing quickly
Interactive AI reply generation	p95 < 8 s; optimize by task/model
Inbox update	< 2 s after normalized event
Message send queue delay	p95 < 5 s for interactive
Workflow execution success	> 99% excluding external provider failures
API availability	99.9% initial target
RPO	≤ 15 minutes for primary business DB

Metric	Initial target
RTO	≤ 4 hours initial; improve for enterprise tier

44.1 Trace IDs

```
trace_id travels through:
HTTP request → domain event → queue job → AI call → provider send → webhook receipt → CRM sync

Support should be able to paste one trace_id and see the full lifecycle.
```

45. Disaster Recovery

- Automated encrypted database backups with point-in-time recovery where supported.
- Daily restore validation in non-production or scheduled restore drills.
- Object storage versioning/lifecycle rules.
- Documented Meta/webhook credential rebuild procedure.
- Runbook for Redis loss: queues restored/replayed from durable DB/outbox where possible.
- Feature flag to pause outbound messaging globally while inbound events continue to ingest.

PART VIII — BILLING, COST & SCALE

How to preserve margins while scaling from early customers to large multi-tenant volume.

46. Subscription and Entitlement Model

Plan dimension	Examples
Users	3 / 10 / 50 / custom
WhatsApp numbers	1 / 3 / 10 / custom
Contacts	5k / 25k / 100k / custom
AI actions	Monthly included actions/credits
Automation runs	Included + overage
Knowledge storage	GB limit
Integrations	SFW only / standard connectors / enterprise API
Features	Target engine, advanced analytics, white label, SSO
Support	Standard / priority / dedicated

47. Usage Metering

Meter usage at the event boundary, not by scanning tables at invoice time.

Metric	Meter event
WhatsApp outbound	message.provider_accepted or billable category event
AI usage	agent_run.completed with provider cost/compute estimate
Automation	workflow_run.started/completed

Metric	Meter event
Contacts	daily peak active contacts
Storage	daily GB snapshot
API	request count for overage plans
Voice/email later	provider usage event

47.1 Cost Ledger

Store internal unit-cost estimates per provider/model/channel and calculate gross margin per tenant. Never hard-code vendor prices into business logic; maintain effective-dated rate cards.

48. SaaS Billing Flow

```
subscription created
→ entitlements cached
→ usage events accumulate
→ threshold alerts at 70/85/100%
→ overage or throttle policy applies
→ invoice generated
→ payment status updates subscription
→ grace period / restricted mode / suspension policy
```

49. Scaling Strategy

49.1 Phase 1 — 0 to ~1,000 Tenants

- One application cluster with horizontal web/worker scaling.
- Managed PostgreSQL primary + backups; Redis managed instance/cluster.
- Separate worker pools by queue.
- pgvector in same PostgreSQL.
- S3-compatible storage + CDN for tenant files.

49.2 Phase 2 — ~1,000 to 10,000 Tenants

- Read replicas for analytics-heavy reads.
- Partition message/event tables.
- Move analytics rollups to dedicated workers/database if needed.
- Dedicated high-throughput webhook/message workers.
- Introduce tenant-aware queue sharding.

49.3 Phase 3 — 10,000+ / Very High Volume

- Shard by tenant group or enterprise dedicated DB.
- Extract channel gateway/AI gateway/analytics if they need independent deployment.
- Adopt durable event bus if Redis/outbox throughput or integration topology justifies it.
- Regional deployment/data residency based on commercial/legal need.

50. AI Cost Optimization

- Rules/classifiers before large-model reasoning.
- Use compact structured memory rather than full transcript.
- Retrieve only top relevant knowledge chunks.
- Cache tenant product/policy summaries.
- Batch embeddings and offline scoring.
- Route simple tasks to smaller/cheaper models.
- Only run message critic where conversion/risk justifies it.
- Dedupe webhook/job retries before invoking AI.
- Cap per-tenant AI spend and expose budget alerts.

PART IX — FASTEST BUILD & LAUNCH PLAN

A practical sequence to reach a sellable production version without overbuilding.

51. MVP Definition

MVP must sell and onboard real tenants. It is not a demo. It needs tenant isolation, WhatsApp connection, consent, inbox, AI qualification/replies, follow-up, human takeover, SFW sync, targets, audit, billing limits and operational monitoring.

51.1 MVP Must-Have Scope

Area	Must ship
Tenancy	Provision tenant/workspace/users/RBAC
WhatsApp	Connect, webhook, send/receive, templates, statuses
CRM	Contacts/leads/deals/stages + CSV import
Consent	Evidence, DND, opt-out, policy gate
Inbox	Conversation list/detail, AI/human modes
AI	Intent, qualification, reply, summary, next best action
Knowledge	Upload + RAG for product/pricing/FAQ
Automation	Follow-up trigger, wait, condition, message, task
Sales	Lead score, hot queue, demo/task
Targets	Daily/weekly/monthly base/stretch + gap
Payments	Adapter + link/status if product requires direct closing
SFW	SSO + lead/event sync + embedded routes
Analytics	Basic funnel/revenue/source/AI usage
Ops	Audit, queue health, retry/dead-letter, feature flags
Billing	Plan entitlements and usage caps

51.2 Defer from MVP

- Full visual workflow marketplace
- Dozens of CRM connectors
- Voice bot
- Complex ML forecasting
- Multi-region active-active
- Dedicated vector database
- Custom agent marketplace
- Advanced white-label/OEM billing
- Fine-tuned custom models

52. Fastest Production Plan

Theme	Deliverables
Foundation	Repo, CI, tenancy, auth/RBAC, canonical CRM schema, environments, audit skeleton
WhatsApp core	Meta app work in parallel; channel account, webhooks, send/status, template sync, diagnostics
CRM + Inbox	Contacts/leads/deals, import, inbox, assignment, message composer, realtime
Consent + AI v1	Consent/preference/suppression, intent, summary, draft reply, knowledge RAG
Sales intelligence	Qualification, lead scoring, next best action, hot queue, human handoff
Automation	Workflow engine v1, follow-up scheduler, retries, dry-run/logs
Targets + Payments	Target engine, forecasts v1, payment adapter/link/recovery events
SFW add-on	SSO, field mapping, two-way lead/activity/payment event integration, embedded navigation
Billing + Analytics	Plans, entitlements, usage, funnel/source/AI performance dashboards
Security + Reliability	Rate limits, key management, retention, DR, idempotency hardening, load tests
Pilot	5–10 controlled tenants, instrumentation, prompt/evaluation tuning, operational runbooks
Launch	Pricing/feature gating, support tools, onboarding docs, monitoring, rollout and sales enablement

52.1 Parallel External Dependency Track

- Meta Business/Tech Provider preparation and App Review/advanced permissions should start in Week 1, not Week 10.
- Set up production payment provider and webhook credentials by Week 4.
- Prepare legal/privacy/terms/consent copy while engineering builds the consent module.
- Pilot customer recruitment starts by Week 6 so real funnel data is available before launch.

53. Suggested Team for Fast Delivery

Role	Count	Focus
Tech Lead / Architect	1	Architecture, code reviews, Meta/SFW integration
Backend Laravel	2	Domain modules, queues, APIs, integrations
Frontend React/Next.js	1–2	Admin, inbox, workflow, dashboards
Flutter	1	SFW mobile surfaces if required in MVP
AI Engineer	1	Gateway, prompts, RAG, evaluations, cost routing
QA Automation	1	API/E2E/load/regression
DevOps	0.5–1	CI/CD, cloud, monitoring, backups
Product/BA	1	Acceptance criteria, workflows, pilot feedback
UI/UX	0.5–1	Design system and critical screens

54. Engineering Standards

54.1 Repository Structure

```
apps/
  api/            Laravel modular monolith
  web/            Next.js tenant + super admin
  sfw-bridge/     optional integration package/service
packages/
  contracts/      OpenAPI/event schemas/TypeScript types
  ui/             shared design system
  crm-sdk/        adapter interfaces + test harness
infra/
  docker/ ci/ terraform-or-equivalent/
docs/
  adr/ api/ runbooks/
```

54.2 Coding Rules

- No direct provider calls from controllers; use domain services/adapters.
- No AI call from view/controller; enqueue or call through AI Gateway.
- No tenant-unscoped repository in tenant business modules.
- Every side-effecting external call has idempotency, timeout and retry policy.
- Every significant configuration is versioned.
- Every workflow/agent action returns structured result codes, not only prose.
- Every public API/event schema has contract tests.

55. Testing Strategy

Layer	Tests
Unit	Scoring, eligibility, target calculations, transforms, state machines
Domain integration	DB constraints, tenancy, outbox, workflow runs
Provider contract	WhatsApp/payment/CRM adapter mocks + sandbox tests
API	Auth, RBAC, validation, idempotency, pagination
E2E	Onboard → inbound chat → qualify → follow-up → payment → won
AI evaluation	Intent accuracy, objection class, hallucination/claim checks, message quality
Security	Tenant isolation, privilege escalation, webhook signature, SSRF/file upload
Load	Webhook bursts, inbox, queue saturation, campaign sends, AI concurrency
Recovery	Retry/dead-letter, provider outage, DB restore, duplicate webhooks

55.1 Golden E2E Scenarios

1. New tenant connects WhatsApp and passes diagnostic.
2. Inbound opted-in customer is qualified and receives AI reply within SLA.
3. Customer requests human; AI pauses and salesperson takes over.
4. No-reply follow-up fires once, then stops when customer replies.
5. Payment link is created, payment webhook marks deal won and cancels recovery.
6. Customer opts out; all marketing sends are blocked immediately.
7. SFW lead update arrives twice; Revenue Cloud processes it once.
8. Tenant A can never access Tenant B via API, queue, search, storage or websocket.
9. AI proposes an unauthorized discount; deterministic validator blocks and creates approval.
10. WhatsApp provider outage causes retry/degraded status without duplicate sends.

56. CI/CD and Environments

Environment	Purpose
Local	Dockerized dependencies, provider mocks
Development	Shared integration, feature branches optional ephemeral
Staging	Production-like, Meta/test credentials, E2E
Pilot/Preprod	Real pilot tenants with restricted volume
Production	Controlled rollout + feature flags

56.1 Pipeline Gates

- Lint/static analysis
- Unit/integration tests
- OpenAPI/event schema compatibility
- DB migration dry run
- Dependency/security scan
- E2E smoke
- Deployment with rollback
- Post-deploy synthetic WhatsApp/inbox health test where feasible

57. Feature Flags and Safe Rollout

Every risky capability—full autonomous sends, negotiation, auto-payment link, new model, new workflow node—must be enableable by internal tenant/plan flag. Pilot on internal/SFW tenants first, then 5%, 25%, 100%.

58. Launch Readiness Checklist

- ☐ Production tenant isolation tests pass
- ☐ Meta app/onboarding path approved/ready for commercial mode
- ☐ Webhook verification/signature/idempotency tested
- ☐ Consent and opt-out flows tested
- ☐ DND enforcement cannot be bypassed by campaign/workflow
- ☐ Human escalation works
- ☐ Plan limits and overages work
- ☐ Audit and trace IDs visible
- ☐ Backups and restore drill complete
- ☐ Queue/dead-letter monitoring configured
- ☐ Provider outages produce degraded mode
- ☐ Payment reconciliation tested
- ☐ SFW two-way sync has conflict policy
- ☐ AI prompts/models versioned and rollback-ready
- ☐ Message templates synced and statuses shown
- ☐ Support diagnostic bundle available
- ☐ Privacy/terms/consent copy approved
- ☐ Pilot funnel and cost metrics reviewed
- ☐ Critical alerts have owners
- ☐ Emergency global outbound pause tested

PART X — PRODUCT SUCCESS & CONTINUOUS IMPROVEMENT

How the SaaS learns, improves conversion and becomes difficult to replace.

59. Analytics Model

59.1 Funnel

```
Eligible Lead → Conversation → Engaged → Qualified → Demo → Quote/Payment Intent → Paid → Onboarded
```

59.2 Segment Every Metric By

- Tenant/workspace
- Industry
- Lead source/campaign
- Language
- Agent profile/version
- Message template/variant
- Sales owner/team
- Deal-size band
- Acquisition cohort
- Channel number

60. Experimentation Engine

A/B testing should optimize collected revenue and qualified conversion. Keep sample allocation, variant, exposure and outcome immutable so experiments remain analyzable.

Test	Primary metric
Opening value proposition	Qualified conversation rate
CTA wording	Demo/payment-intent rate
Follow-up delay	Reply + conversion rate
Proof type	Objection-to-demo conversion
AI tone/language	Qualified and won rate
Demo slot choice	Booking rate
Payment recovery message	Recovered payment rate

61. AI Evaluation Loop

1. Sample production conversations with outcome labels.
2. Create evaluation sets by industry/intent/language.
3. Score classification accuracy, factuality, prohibited claims, repetition and conversion proxy.
4. Compare prompt/model versions offline.
5. Canary new version on limited tenants.
6. Measure actual business outcomes and cost.
7. Promote or roll back with version control.

62. Industry Template Packs

• SaaS/software

• Real estate

• FMCG/distribution

• Automobile/dealers

• Education/coaching

• Travel/hospitality

• Professional services

• Retail

• Home services

• Fitness/wellness

• Restaurants/food

• Clinics/dental subject to channel/legal restrictions

Each pack should define terminology, qualification fields, pipeline stages, common objections, value propositions, knowledge starter set, workflow templates and prohibited/regulated behavior. Do not assume one script fits all industries.

63. Agent / Workflow Marketplace Roadmap

- Installable industry agent packs
- Approved objection libraries
- Revenue recovery packs
- Renewal/upsell agents
- Collections agents where lawful and policy-compliant
- Partner-published connectors
- Template packs with ratings/version compatibility

Marketplace is Phase 2/3. Build internal package/version interfaces now so later distribution does not require rewriting configs.

64. Customer Success Automation

- Onboarding checklist and health score
- Detect disconnected WhatsApp/failed webhooks
- Detect no knowledge source / stale pricing
- Detect AI disabled or unassigned hot leads
- Weekly ROI report: leads handled, demos, revenue influenced, recovered payments
- Renewal risk alerts and plan-usage recommendations
- Feature adoption nudges tied to measurable outcomes

65. Product Health Score

```
tenant_health = channel_health + integration_health + automation_success + AI_quality +
                user_adoption + funnel_activity + billing_status

Use health score to drive support outreach before churn occurs.
```

66. Risk Register

Risk	Severity	Mitigation
Meta onboarding/app approval delay	High	Start immediately; test mode + external dependency track
Unsolicited messaging / account restrictions	Critical	Consent center, policy gate, frequency/quality monitoring
Google Maps scraping dependency	Critical	Exclude from product; licensed/permitted data only
AI hallucinated pricing/claims	High	Structured product rules + RAG + validator + approvals
Cross-tenant data leak	Critical	Central tenant scope, tests, storage/cache isolation
Duplicate sends/payment links	High	Idempotency + provider refs + locks
Queue backlog during campaigns	High	Separate worker pools + throttling
AI cost explosion	Medium/High	Model routing, budgets, compact memory, caching

Risk	Severity	Mitigation
CRM sync conflicts	Medium	Field source-of-truth + mapping version + dead-letter
Low conversion despite replies	Business	Optimize revenue metrics; experiment and target best segments
Over-automation annoys users	High	Autonomy modes, stop rules, human handoff
Single AI vendor outage	Medium	Provider abstraction/fallback where commercially justified

67. Future Roadmap

Phase	Potential capability
After PMF	Email/SMS/voice conversation adapters
After PMF	Advanced forecasting and cohort models
After PMF	Partner/OEM marketplace
Scale	Dedicated tenant shards and region deployment
Scale	Event-stream analytics warehouse
Scale	More CRM connectors and connector SDK portal
Advanced AI	Tool-use policies by regulated vertical
Advanced AI	Tenant-specific evaluation/tuning pipelines
Revenue expansion	Upsell, renewal, referrals, customer-success agents

PART XI — IMPLEMENTATION REFERENCE

Acceptance criteria, sample algorithms, source links and developer handoff notes.

68. Core Acceptance Criteria by Module

Module	Definition of Done
Tenancy	No tenant business query runs without tenant context; automated isolation test suite passes.
WhatsApp	Inbound/outbound/status webhooks are idempotent and linked to tenant/channel account.
Consent	Outbound marketing is blocked when consent/suppression rules fail, regardless of workflow or user.
Inbox	Human takeover stops AI sends immediately; returning to AI resumes only from explicit state.
AI	Every production run stores task/model/prompt version/usage/structured outcome.
Knowledge	Retrieval can never cross tenant/workspace and respects active knowledge versions.
Automation	Published workflow version is immutable; retries do not duplicate side effects.
Targets	System calculates remaining base/stretch gap and required activity from measured conversion rates.
CRM	Adapter mapping and source-of-truth rules prevent endless update loops.
Payments	Paid status stops recovery and is reconciled with provider.
Billing	Entitlements are enforced server-side, not only hidden in UI.
Audit	Admin can trace a deal-winning conversation from lead source through messages, AI decisions and payment.

69. Sample Orchestrator Pseudocode

```
onInboundMessage(message):
  ctx = loadConversationContext(message)
  if ctx.isSuppressedForAllProcessingRequiredByPolicy():
      persistOnlyAndRouteAsRequired()
      return

  intent = AI.classifyIntent(message, ctx)
  memory = Memory.update(ctx.memory, message, intent)
  leadState = SalesState.update(ctx.lead, intent, memory)

  if intent.requiresHuman() or leadState.value > tenant.autoHumanThreshold:
      Handoff.assignBestUser(ctx)
      return

  action = NextBestAction.choose(leadState, ctx.goals, ctx.policies)
  if action.type == SEND_MESSAGE:
      draft = AI.draft(action, ctx, Knowledge.retrieve(action, ctx))
      checked = MessageQuality.validate(draft, ctx)
      eligibility = PolicyGate.canSend(ctx.contact, checked.intent, ctx.channel)
      if eligibility.allowed:
          MessageService.sendIdempotently(checked)
      else:
          createBlockedAction(eligibility.reason)
  else:
      ActionExecutor.execute(action)

  RevenueTargetEngine.enqueueRecalculation(ctx.tenant_id)
```

70. Sample Target Recalculation Pseudocode

```
recalculateGoal(goal):
  actual = Metric.value(goal.scope, goal.period)
```

```
  remaining = max(goal.stretch_target - actual, 0)
  days = remainingWorkingDays(goal.period, goal.timezone)
  rates = FunnelRates.forScope(goal.scope).withConfidenceFloor()

  requirements = reverseFunnel(remaining, rates)
  pipeline = OpportunityQueue.rank(goal.scope)
  forecast = Forecast.from(pipeline, rates, upcomingActivities())

  snapshot = GoalSnapshot(actual, remaining, requirements, forecast)
  save(snapshot)

  if forecast.expected < goal.base_target:
      emit('goal.at_risk', snapshot)
  return snapshot
```

71. Sample Message Eligibility Pseudocode

```
checkOutbound(contact, account, message):
  deny if tenant/account inactive
  deny if contact.dnd or suppression active
  deny if purpose/category lacks required eligibility/consent
  deny if quiet hours and not permitted urgent utility case
  deny if frequency cap exceeded
  deny if vertical/geography policy blocks message

  if now <= conversation.service_window_until:
      require message type allowed in service window
  else:
      require approved template and intended category match

  return allow
```

72. Pilot Tenant Playbook

1. Select 5–10 businesses with clear sales funnels and responsive owners/managers.
2. Import only permitted leads and document consent/source states.
3. Run Assist mode for first days to collect good/bad reply examples.
4. Enable AI automation for low-risk qualification and follow-up first.
5. Keep negotiation/discount human-approved until performance is measured.
6. Review daily funnel: eligible → conversation → qualified → demo → paid.
7. Tag every lost reason and message complaint/opt-out.
8. Iterate playbooks weekly; do not optimize only reply rate.
9. Publish case studies only with customer approval and verifiable outcomes.

73. Launch KPI Dashboard

KPI	Launch review question
Tenant activation rate	Can companies connect WhatsApp and go live without engineering help?
Time-to-first-conversation	How quickly from signup to useful automation?
AI acceptance rate	How often do humans send suggested replies unchanged?
Human handoff rate	Where is AI still weak or intentionally constrained?
Opt-out/block signals	Are cadences/message value healthy?
Qualified conversion	Are we attracting and identifying the right leads?
Payment recovery	Is the engine preventing late-funnel leakage?
Gross margin	Are AI/channel costs aligned with plan pricing?
Support tickets per tenant	What is still operationally confusing?

KPI	Launch review question
Churn / retention	Does the platform prove ongoing revenue value?

74. Official Reference Sources

WhatsApp Business Messaging Policy: https://whatsappbusiness.com/policy/ — Opt-in, templates, 24-hour service window, escalation, quality/enforcement.

Meta official WhatsApp Business Platform Postman workspace: https://www.postman.com/meta/whatsapp-business-platform/ — Official API examples including Cloud API and Embedded Signup collections.

Google Maps Platform Terms: https://cloud.google.com/maps-platform/terms — Current restrictions include no scraping/export of Google Maps Content for outside use.

Digital Personal Data Protection Act, 2023: https://www.meity.gov.in/static/uploads/2024/02/Digital-Personal-Data-Protection-Act-2023.pdf — India personal-data framework including consent and withdrawal.

Digital Personal Data Protection Rules, 2025: https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa — Final rules and phased commencement information.

Important: API versions, commercial pricing, message categories, onboarding requirements and regulatory obligations can change. Store provider rate cards and policy capability flags as configuration, and re-check official documentation before each production release.

75. Final Developer Directive

Build one reusable, tenant-isolated Revenue Automation Cloud. Treat SFW as the first deeply integrated distribution channel—not as the platform boundary. Deterministic code owns permissions, consent, state, pricing limits, billing and irreversible actions. AI owns language, classification, summarization and constrained sales reasoning. Instrument every step so the product can learn from real revenue outcomes.

The fastest route to a world-class product is disciplined sequencing: first make onboarding, messaging, consent, inbox, AI assistance, follow-up and SFW synchronization exceptionally reliable; then add autonomous negotiation, additional channels, marketplaces and high-scale service decomposition after real customers prove the need.

PART XII — V2 AUTONOMOUS OMNICHANNEL GROWTH EXPANSION

One-click onboarding • AI chatbot • Bulk campaigns • RCS • SMS • Self-marketing • Reseller billing

76. Version 2.0 Product Mandate

Build one core AI Revenue & Growth OS once. Sell it as standalone SaaS, embed it inside SFW, expose it through APIs to any CRM, and allow white-label/reseller distribution. WhatsApp is the primary channel, but the product architecture must be channel-agnostic.

The differentiator is not “bulk messaging.” The product must behave as an autonomous revenue operator that understands a company, creates compliant campaigns and chatbots, connects communication channels, converts replies into sales conversations, routes high-value work to humans, attributes revenue, and continuously improves against business targets.

Product principle	Engineering implication	Reason
Goal-driven automation	Every campaign/workflow begins with an objective and measurable KPI	Prevents activity without business outcome
AI proposes; deterministic controls govern	Policy, consent, billing, permissions, money and state transitions remain coded rules	AI must never bypass hard constraints
Easy enough for non-technical SMBs	Wizard-first UX plus AI setup from website/docs	Reduces implementation/support cost
Enterprise-safe	RBAC, approvals, audit logs, tenant isolation, SSO, data controls	Enables larger customers
Provider-agnostic	Adapters for Meta direct, BSP, RCS provider, SMS aggregator	Avoids vendor lock-in
Revenue optimization	Optimize qualified leads, demos, payments and revenue—not message volume	Creates measurable customer ROI

77. Pain-Point-to-Solution Matrix

Customer pain point	Product solution	Automation level
WhatsApp API onboarding is confusing	Embedded Signup / provider connect wizard, automatic webhooks, template sync, health test	High
Business owner cannot build chatbot	AI One-Minute Chatbot Wizard from website, documents or plain-language description	Very high
Templates get rejected or perform poorly	AI Template Studio with policy linting, variable checks, rejection feedback loop and versioning	High; approval cannot be guaranteed

Customer pain point	Product solution	Automation level
Bulk campaigns feel like blasting	Eligibility, consent, segmentation, send controls, A/B testing, reply routing and revenue attribution	Very high
Follow-ups are forgotten	Event-driven smart follow-ups based on lead state and buying intent	Very high
Sales team wastes time on cold leads	Intent, fit, engagement and purchase-probability scoring	High
Channels are fragmented	Unified WhatsApp/RCS/SMS conversation and campaign layer	High
Marketing team lacks content	AI Growth Planner + campaign calendar + copy/template/offer generator	High
Provider costs are difficult to control	Rate cards, wallet, top-ups, usage ledger, margin rules and alerts	Deterministic
CRM data becomes stale	Two-way event/webhook synchronization	High
Owner does not know what works	End-to-end attribution from campaign to payment	High
Agency/reseller cannot manage clients	Reseller hierarchy, sub-tenant provisioning, markup, wallet and white-label	High

78. One-Click WhatsApp SaaS Connection

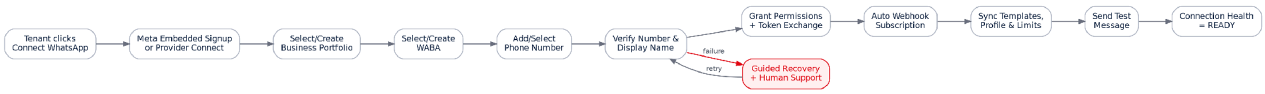

Figure 12 — Target WhatsApp connection experience: guided, recoverable and testable.

Preferred implementation: Meta Embedded Signup when the SaaS is approved/eligible for the required platform role. Keep a second provider-adapter path so a tenant can connect through an approved BSP/aggregator. Never design the primary UX around asking customers to paste permanent access tokens manually.

1. Click “Connect WhatsApp”. Ask whether the tenant wants Meta Direct or an available provider route.
2. Launch the provider/Meta authorization window and capture the resulting business, WABA and phone-number identifiers.
3. Create/update the tenant channel_account atomically; encrypt secrets and store provider-scoped credentials in the secrets service.
4. Automatically subscribe webhooks and validate the callback signature/challenge.
5. Sync display profile, templates, message capabilities, quality/health metadata and limits exposed by the provider.
6. Run a connection diagnostic and optional test-message flow.
7. Mark the channel READY only when send and webhook checks pass. Otherwise show a guided recovery state with actionable error text.
8. Continuously monitor authentication expiry, webhook health, phone status, template status and provider outages.

Brand name, logo/profile appearance and verification are subject to Meta/provider rules and approval. The SaaS must describe this as an onboarding capability—not a guaranteed approval outcome.

79. WhatsApp Connection State Machine & Recovery

State	Meaning	Allowed actions
NOT_CONNECTED	No usable account exists	Start connection
AUTHORIZING	OAuth/embedded signup in progress	Cancel/retry
CONFIGURING	Saving IDs, permissions, webhooks	Read-only progress
VERIFYING	Connection and webhook diagnostics	Retry diagnostics

State	Meaning	Allowed actions
READY	Channel can send/receive subject to policy	Campaigns, inbox, templates
DEGRADED	Partial capability or webhook/provider issue	Receive alerts; limited sending if safe
ACTION_REQUIRED	Tenant must resolve business/number/template issue	Guided fix
SUSPENDED	Provider/platform restriction or admin stop	No marketing sends
DISCONNECTED	Credentials revoked / explicit disconnect	Reconnect

- Every provider error must map to a normalized error code and a human-readable remediation message.
- Never silently fall back to another tenant phone number.
- Connection credentials must be tenant-scoped, encrypted and never returned in normal API responses.
- Webhook subscriptions and channel configuration must be idempotent so retries are safe.

80. AI One-Minute Chatbot Builder

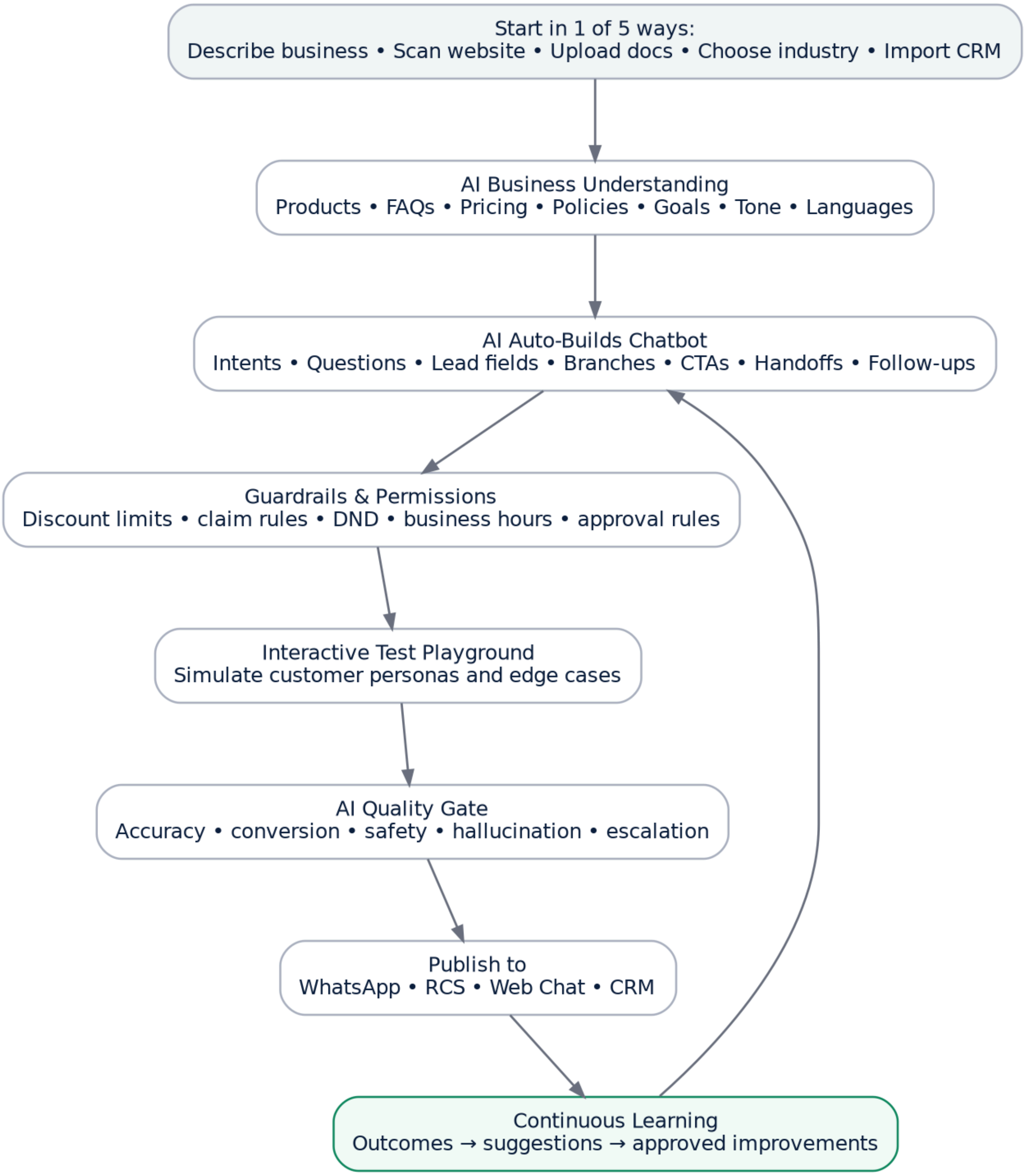

Figure 13 — AI-first chatbot creation flow for non-technical users.

The chatbot builder must support five starting methods: (1) describe the business in plain language, (2) scan/import an allowed website, (3) upload documents/catalogues/FAQs, (4) choose an industry template, or (5) import product/service and CRM fields. The user should be able to publish a useful first bot without drawing a flowchart.

Wizard step	What user does	What AI does automatically
1. Business	Name, industry, website/docs	Extract business summary, products, services, locations, FAQs
2. Objective	Choose lead generation, sales, support, bookings, collections etc.	Select starter intents, fields, actions and KPIs
3. Tone & Language	Choose languages/tone or accept recommendation	Create language policy, examples and fallback style

Wizard step	What user does	What AI does automatically
4. Commercial Rules	Confirm pricing, discount limits, hours, escalation	Create hard guardrails and approval matrix
5. Preview	Chat with bot as a customer	Run test personas, identify missing answers and weak branches
6. Publish	Choose channels	Create version, activate webhooks/routing and monitor

81. Chatbot Runtime Rules

Rule	Required behavior
Answer grounding	Use tenant knowledge and structured product/pricing data; do not invent unavailable facts.
Intent handling	Classify intent and choose a deterministic action policy before language generation.
Lead capture	Ask only the minimum fields required for the current objective; progressively enrich later.
Pricing	Read current price/offer from product tables. Never create an unapproved discount.
Sensitive/regulated flows	Use tenant-configured restricted-flow policies and mandatory human handoff where applicable.
Uncertainty	If confidence is below threshold, ask a concise clarification or escalate.
Human takeover	Stop autonomous sales replies when a human locks the conversation; resume only by explicit release rule.
Opt-out	Process opt-out immediately and update suppression state before any other marketing action.
Memory	Persist structured facts separately from generated summaries; summaries are not the canonical source.
Versioning	Every published bot has immutable version metadata; rollback must be one click.

```
on_inbound_message(event):
  tenant = resolve_tenant(event.channel_account_id)
  conversation = upsert_conversation(event)
  persist_raw_event(event)
  if human_lock(conversation): notify_human(); return
  intent, confidence = classify_intent(event.text, context)
  action = policy_engine.decide(intent, conversation, tenant_rules)
  if action.requires_human: handoff(); return
  facts = retrieve_grounded_context(action)
  draft = ai_generate(action, facts, conversation_memory)
  final = safety_and_quality_gate(draft, action, tenant_rules)
  execute(final.action)
  persist_decision_trace()
```

82. Natural-Language Automation Builder

In addition to a visual flow builder, allow users to type rules such as: “If a customer asks price, show the current plans. If they ask for a discount above 5%, assign the sales manager. If no reply for 3 days, send the approved follow-up template if eligible.” AI converts this into a draft workflow graph.

Stage	System behavior
Parse	Convert natural language into trigger, conditions, actions, delay and exception branches.

Stage	System behavior
Validate	Check referenced fields, users, templates, channels, permissions and circular loops.
Simulate	Run sample events and show exactly what would happen.
Approve	User confirms draft; high-risk actions require explicit admin approval.
Publish	Create immutable workflow version and activate.
Observe	Show runs, failures, conversion and AI recommendations.

AI can generate workflow drafts, but it must never directly publish a money-moving, discounting, data-deleting or mass-marketing workflow without the configured approval policy.

83. AI Template Studio for Meta Approval

The AI Template Studio should optimize template quality and approval likelihood, but the product must never claim it can guarantee Meta approval. Approval is an external platform decision. The system should learn from tenant-specific approvals/rejections and improve future drafts.

Input	AI behavior
Campaign objective	Choose appropriate template intent/category candidates and CTA strategy.
Audience/industry	Use relevant vocabulary and avoid misleading personalization.
Offer/product	Pull only approved offer facts and current pricing.
Language	Generate compliant localized variants.
Template history	Prefer structures with strong approval and conversion history.
Policy rules	Flag risky claims, prohibited content, missing context or invalid variables/buttons.
Rejection reason	Explain the issue, generate corrected versions and preserve version history.

Template quality dimension	Example checks
Policy lint	Prohibited claims, aggressive/spam-like language, unsupported content patterns
Variable lint	All variables have examples; no ambiguous/unbounded substitutions
Clarity	Business identity and message purpose are understandable
CTA relevance	CTA matches campaign goal and does not create deceptive urgency
Localization	Natural language, correct placeholders and cultural tone
Conversion quality	One primary action, concise message, useful value proposition
Approval evidence	Prior approval/rejection patterns stored by provider, category and locale

84. Template Lifecycle

```
DRAFT -> AI_REVIEWED -> HUMAN_APPROVED(optional) -> SUBMITTED
SUBMITTED -> APPROVED | REJECTED | PAUSED | DISABLED
REJECTED -> AI_REMEDIATION_DRAFT -> REVIEW -> RESUBMIT
APPROVED -> VERSION_LOCKED -> AVAILABLE_FOR_CAMPAIGNS
```

- Never edit an approved production template “in place” in the SaaS model; create a new local version when content changes.
- Sync provider template statuses by webhook/polling reconciliation.
- Campaigns must reference exact template version and locale.
- If a production template becomes unavailable, automatically pause affected outbound jobs and alert the tenant.

85. Bulk WhatsApp Campaign Engine

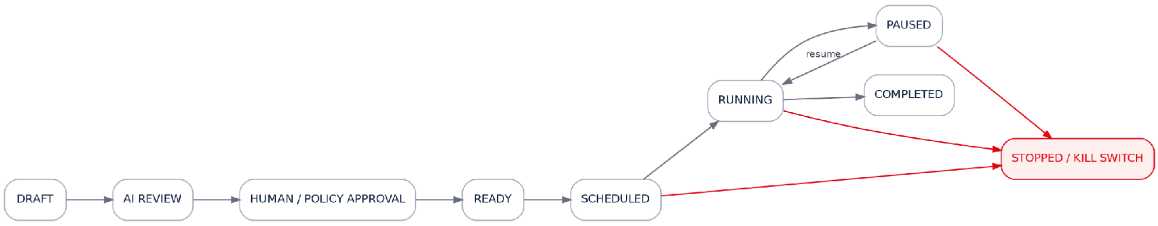

Figure 14 — Campaign execution state machine and emergency stop path.

Bulk WhatsApp marketing must be implemented as controlled campaign orchestration. Every recipient passes the same eligibility guard used by one-to-one automation before a send job is created.

Pre-send gate	Required checks
Identity	Correct tenant, channel account and recipient
Consent	Current marketing eligibility/opt-in evidence according to configured policy
Suppression	DND, global/tenant/channel opt-out, legal suppression
Frequency	Tenant and platform frequency caps, recent campaign history
Template	Approved/available template and correct locale
Data quality	Required variables exist and are safe to substitute
Account health	Channel READY and not paused/restricted by internal health rules
Budget	Wallet/credit/entitlement permits the send
Schedule	Allowed tenant/recipient time window
Idempotency	Recipient + campaign + version has not already been sent unless retry policy allows

```
eligible = consent_ok
      and not suppressed
      and frequency_ok
      and template_ready
      and channel_healthy
      and billing_authorized
      and schedule_ok
      and idempotency_ok
if not eligible: store_skip_reason(); do_not_enqueue()
```

86. Autonomous Self-Marketing Studio

Figure 15 — Goal-to-revenue marketing autopilot.

The self-marketing layer should let a tenant specify a business target instead of manually creating every campaign. Example: “Generate 150 qualified leads next month for our dental clinic plan.” The Growth Planner reverse-calculates the required funnel, builds a proposed campaign calendar, generates assets and requests only the approvals needed by policy.

AI marketing capability	Output
Growth goal planner	Revenue/lead/demo target translated into weekly and daily activity requirements
Campaign calendar	Recommended campaigns by lifecycle stage, segment, channel and date
Offer assistant	Offer framing using tenant-approved products, margins and discount rules
Audience discovery	Segments from CRM behavior, lifecycle, location, product, intent and engagement
WhatsApp template generator	Policy-aware campaign template variants
RCS creative composer	Text + rich card/carousel structures + suggested actions
SMS composer	DLT-template-aware SMS copy and fallback variants
Landing-page copy	Headline, proof, CTA, form fields, UTM campaign mapping
QR / click-to-chat tools	Campaign QR and tracked deep-link generation
Referral engine	Referral messages, unique links/codes, reward workflows
Reactivation engine	Dormant lead/customer win-back campaigns
Renewal/upsell engine	Lifecycle-based offers and reminders
Social/content planner	Optional content calendar and channel-ready copy via pluggable publishing connectors
Performance optimizer	Shift future traffic toward variants that produce qualified revenue

87. AI Growth Planner Rules

- Never create campaign volume solely to hit a message-count target. Re-prioritize payment-pending, hot, demo-completed and warm opportunities before acquiring more cold activity.
- Use actual tenant funnel conversion rates after sufficient data exists. Until then, label assumptions and maintain conservative ranges.
- External ad-spend changes require approval above tenant-configured thresholds. AI may recommend budgets; deterministic rules govern spending authority.
- Campaign suggestions must include expected objective, audience, channel, offer, CTA, required approval, estimated cost range and measurable success criteria.
- Every AI recommendation must be dismissible, explainable and logged.

88. Omnichannel Orchestration

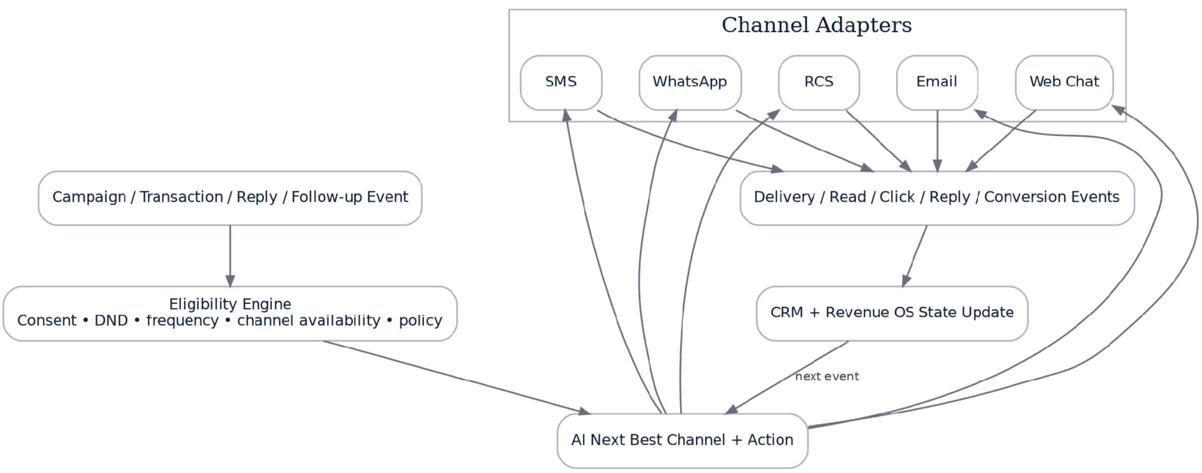

Figure 16 — One decision layer, multiple communication adapters.

Implement a generic ChannelAdapter contract so conversation, campaign, delivery and billing logic does not depend on a specific provider. WhatsApp, RCS, SMS and email adapters expose different capabilities but share a normalized event model.

```
interface ChannelAdapter {
  connect(tenant, auth_payload)
  disconnect(channel_account)
  capabilities(channel_account)
  send(message_command)
  send_template(template_command)
  upload_media(media)
  normalize_webhook(raw_event)
  sync_templates_or_assets()
  health(channel_account)
  cost_hint(message_command)
}
```

Routing input	Example influence
Customer preference	Prefer opted-in/preferred channel
Channel capability	RCS device capability, WhatsApp account readiness
Message purpose	Marketing vs transaction vs OTP vs support
Urgency	SMS fallback may be preferred for critical transactional notice
Cost	Use provider rate card within quality constraints
Engagement history	Prefer channel with stronger recent response
Policy/consent	Block channels lacking required permission
Content richness	RCS for rich carousel; WhatsApp for conversation; SMS for concise fallback

89. RCS Messaging Module

RCS is implemented as a first-class adapter, not a separate disconnected application. Support agent/brand onboarding state, webhook events, capability checks, text/media, rich cards, carousels, suggested replies/actions, delivery events and fallback policies. Google RCS for Business currently supports rich cards, media and interactive suggestions; device/client capabilities can differ, so the adapter must check capability before selecting rich content.

RCS feature	Implementation requirement
Agent/brand profile	Store provider agent ID, brand, launch/verification state and regions
Capability check	Cache short-lived recipient capability result and respect provider guidance
Rich card	Media, title, description and suggested actions/replies
Carousel	2–10 cards where provider capability permits; reusable asset model
CTA actions	Open URL, dial, location/calendar actions where supported
Delivery events	Normalize sent/delivered/read/reply/failure into channel events
Expiration	Support message TTL/expiry for time-sensitive content
Fallback	Route to SMS/other channel only when policy and tenant rule allow
Billing	Record provider billing event and map to effective rate card

90. Bulk SMS Module for India (DLT-Aware)

For India bulk SMS, build DLT onboarding and enforcement into the tenant setup. TRAI guidance requires senders/principal entities using bulk communication to complete the applicable registrations, including Principal Entity registration, headers and content templates; consent templates/consumer consent may also apply depending on the communication.

DLT object	Tenant setup fields
Principal Entity	PE name, PE ID, registration status, operator/provider, evidence/document references
Header/Sender ID	Header, purpose/type, approval status, effective date
Content Template	Template ID, category, exact fixed/variable structure, status, locale
Consent Template	Consent template ID and status when applicable
Provider Mapping	SMS aggregator account, route, credentials, callback configuration
Audit	Submission/update history and responsible user

Do not allow an India bulk-SMS campaign to send unless required DLT identifiers/templates are mapped and the provider validation passes. DLT is a workflow constraint, not a help-page note.

91. Channel Fallback Strategy

Scenario	Default decision pattern
WhatsApp marketing not eligible	Do not auto-switch to SMS unless SMS consent/regulatory rules independently permit it.
RCS unsupported device	Use tenant-approved SMS fallback template if eligible.
RCS rich media unsupported	Downgrade to supported RCS text/card variant before cross-channel fallback.
Transactional urgent notice	Use configured priority chain; record why each channel was attempted/skipped.
Customer replies on one channel	Prefer continuing that active conversation channel unless business rule requires otherwise.
Channel outage	Pause marketing; for critical transaction events use approved fallback policy only.

92. Unified Campaign & Conversation Attribution

```
growth_plan_id
 -> campaign_id
  -> segment_id
   -> campaign_recipient_id
    -> outbound_message_id
     -> conversation_id
      -> lead/opportunity_id
       -> appointment/quotation/payment_id
        -> recognized_revenue
```

- Every click uses a tenant-safe tracked-link ID with campaign/message attribution.
- QR codes resolve to tracked campaign destinations and never expose internal IDs.
- Multi-touch attribution can be reported, but preserve a deterministic primary attribution rule for finance/reconciliation.
- Campaign optimization uses qualified revenue and gross margin when available, not only read/click/reply rate.

93. Live Agent Support & Human-AI Collaboration

Mode	Behavior	Best for
Autonomous	AI handles permitted conversation until handoff trigger	SMBs, high-volume qualification
Hybrid	AI qualifies/nurtures; human owns negotiation/close	B2B, SFW sales teams

Mode	Behavior	Best for
Copilot	AI drafts and recommends; human approves each send	Enterprise / sensitive teams
Human lock	Assigned agent exclusively controls thread until release	Complex/angry/VIP customers

- User Management and RBAC are required: tenant admin, manager, campaign operator, sales agent, support agent, viewer and API user.
- Assignment rules must support round robin, skill/language, geography, product, team capacity, lead score and VIP routing.
- AI should generate a handoff summary, lead score, intent, objections, promised actions and recommended next step.
- Delivery/channel charges remain metered independently from human seat/licence fees.

94. URL Tracking, QR and Lead Capture Tools

Tool	Purpose	Core data
Tracked links	Measure CTA click and downstream conversion	link_id, campaign_id, destination, UTM, clicks, unique clicks
QR generator	Offline-to-chat/landing-page conversion	qr_id, campaign, destination, scans, source
Click-to-WhatsApp links	Drive customer-initiated conversations	campaign/referral params, prefilled context
Landing-page forms	Consent/lead capture	form version, purpose, fields, consent evidence
Website chat widget	Inbound lead capture and chatbot channel	site, page, visitor session, conversation
Referral links/codes	Attribute advocates and rewards	referrer, referred lead, reward state

95. Provider Abstraction, Rate Cards, Wallets & Billing

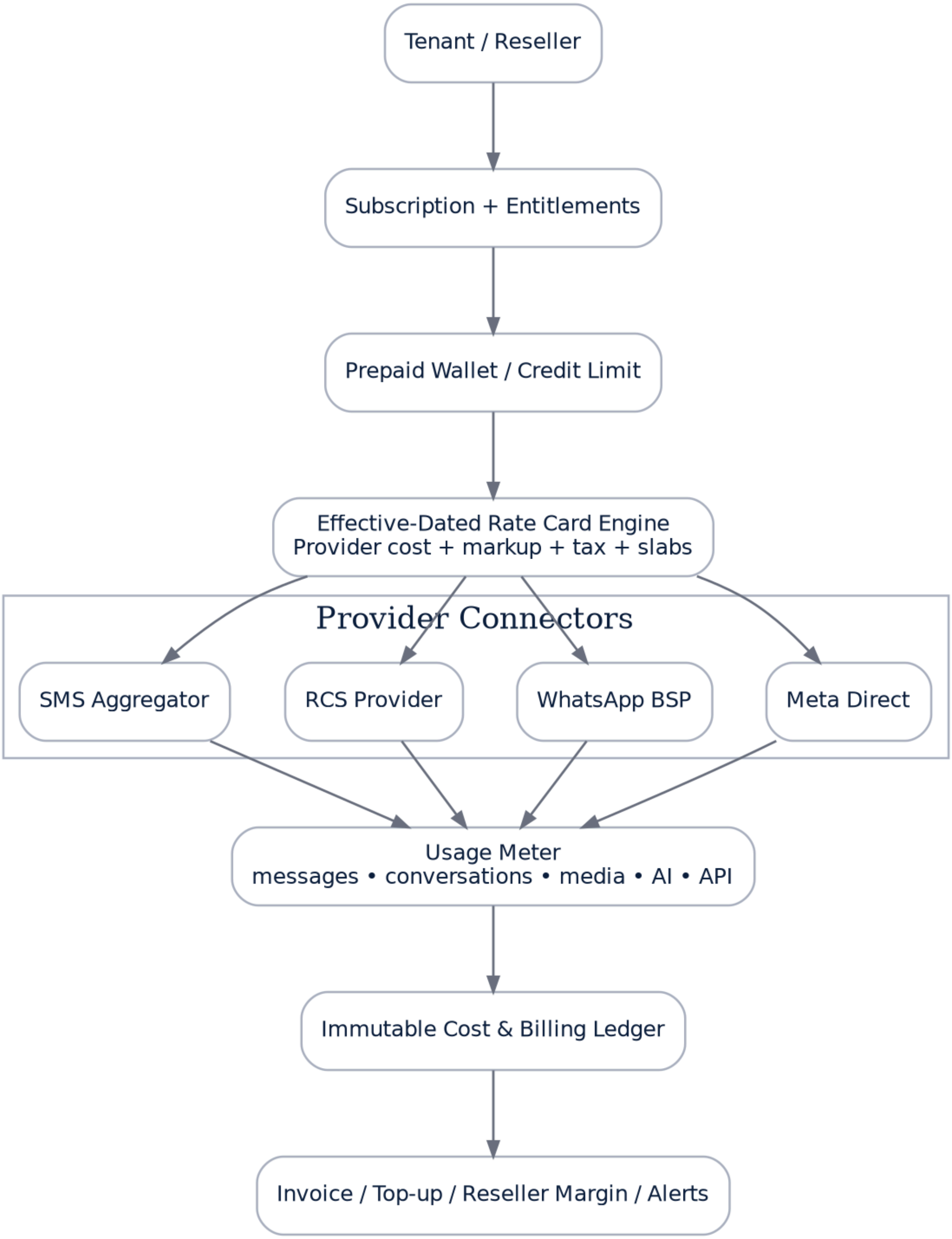

Figure 17 — Provider-independent commercial and usage architecture.

Never hard-code provider commercials in source code. Create an effective-dated rate-card engine capable of provider cost, platform markup, reseller markup, tax, volume slabs, minimum fees and free/zero-rated events. Preserve the raw provider billing event for reconciliation.

Entity	Key fields
provider	type, name, status, credentials strategy, settlement terms
rate_card	provider_id, currency, valid_from/to, tenant/reseller scope, priority

Entity	Key fields
rate_card_item	channel, message_type/category, country/region, unit, slab_from/to, provider_cost, sell_rate, tax_rule
wallet	tenant/reseller, currency, balance, credit_limit, auto_topup settings
wallet_transaction	debit/credit, reason, provider event, idempotency key, balance_after
usage_event	channel, provider, tenant, message/campaign, billable unit, provider cost estimate
invoice_line	usage aggregation, subscription fee, tax, discount, reseller margin

96. Commercial Rate Card Supplied by Product Owner — Configurable Example

The following figures are commercial inputs supplied for planning. They are NOT to be represented in code or UI as permanent official Meta/Google/TRAI tariffs. Store them as provider rate-card records with effective dates and allow admin updates without deployment.

Channel / item	Supplied commercial input	Developer treatment
WhatsApp Marketing	“0.87 paisa + GST”	Unit is ambiguous: confirm whether supplier means ₹0.87 (87 paise) or ₹0.0087 before production billing.
WhatsApp Utility	“0.14 paisa + GST”	Same unit confirmation required; rate-card item.
WhatsApp Authentication	“0.14 paisa + GST”	Same unit confirmation required; rate-card item.
User-initiated WhatsApp	“Free conversation”	Provider-specific commercial rule; never infer official platform pricing.
WhatsApp setup	Free of cost	Provider onboarding fee item = 0 if contract confirms.
Monthly API fee	₹1,000	Subscription/add-on fee, effective-dated.
AI chatbot	As per requirement	Product plan/add-on pricing, separate from channel cost.
RCS 1 lakh	₹0.12 + GST; unlimited validity	Volume slab/rate-card record.
RCS setup / annual	Free	Provider rate-card entries.
Bulk SMS 50,000	₹0.13 + GST; unlimited validity	Volume slab.
Bulk SMS 1 lakh	₹0.12 + GST; unlimited validity	Volume slab.
Bulk SMS 2 lakh	₹0.11 + GST; unlimited validity	Volume slab.

The provider feature set supplied for planning—brand profile/logo capability where approved, live-agent support, user management, chatbot integration, full API access, access controls, URL tracking, top-up, reseller features, CRM integration, delivery reports, RCS CTA/media, and SMS onboarding assistance—should be expressed as provider capabilities and product entitlements rather than assumptions baked into the core.

97. Reseller / White-Label Operating Model

```
Platform Super Admin
  -> Reseller
      -> Reseller Admin Users
      -> Wallet / Credit / Markup Rules
      -> Client Tenant A
      -> Client Tenant B
      -> Client Tenant C
```

Reseller feature	Requirement
Provisioning	Create/disable client tenants within reseller entitlement
Markup	Per-channel and subscription markup with floor/ceiling controls
Wallet	Parent-funded or client-funded wallet mode
White-label	Logo, domain, colors, email sender and support details
Reporting	Client usage, revenue, margin, wallet exposure and failed payments
Access	Reseller support access must be explicit, auditable and tenant-approved where required
API	Scoped reseller API for provisioning, balances and usage
Commercial privacy	Client never sees raw upstream provider cost unless reseller chooses to expose it

98. Autonomous Campaign Optimizer

The optimizer should be conservative early and data-driven later. Use staged experimentation: deterministic A/B tests first; optional multi-armed-bandit allocation only after minimum sample and guardrail thresholds are reached.

Optimization target	Primary metric	Guardrail
Acquisition	Qualified lead cost	Opt-out/complaint/quality health
Qualification	Qualified rate	False-positive human review rate
Demo	Demo booked/completed rate	No-show rate
Closing	Paid conversion / gross margin	Discount rate and refunds
Campaign	Revenue per eligible recipient	Message cost and negative feedback
Channel routing	Revenue or task completion per cost	Consent, deliverability, latency
Template selection	Approved + converted revenue	Approval status and quality signals

99. AI Learning & Governance Loop

1. Capture outcome signals: delivered, read, click, reply, intent, qualified, demo, quote, payment, refund, opt-out and human corrections.
2. Aggregate by tenant, industry, campaign objective, locale, template version and agent version.
3. Generate improvement recommendations; do not silently rewrite production prompts/workflows.
4. Run offline evaluation against a tenant-safe test set and regression scenarios.
5. Require approval for changes that affect customer-facing behavior unless tenant explicitly enables bounded auto-optimization.
6. Publish a new version with experiment allocation and rollback.
7. Monitor for degradation and automatically roll back when configured safety thresholds are breached.

100. Additional V2 Data Model

Table family	Key tables / purpose
Channel/provider	providers, provider_connections, channel_accounts, channel_capabilities, channel_health_events
Chatbot	chatbot_projects, chatbot_versions, chatbot_intents, chatbot_actions, chatbot_test_cases, chatbot_evaluations
Templates	message_templates, template_versions, template_submissions, template_provider_status_events, template_quality_scores
Growth	growth_plans, growth_targets, marketing_calendar_items, ai_recommendations, approval_requests

Table family	Key tables / purpose
Campaign	campaign_assets, campaign_experiments, campaign_variants, recipient_eligibility_snapshots, send_batches
RCS	rcs_agents, rcs_assets, rcs_capability_checks, rcs_message_events
SMS/DLT	sms_dlt_profiles, sms_headers, sms_content_templates, sms_consent_templates
Tracking	tracked_links, tracked_link_events, qr_codes, qr_scan_events, landing_pages, form_submissions
Commercial	rate_cards, rate_card_items, wallets, wallet_transactions, usage_events, provider_billing_events, reseller_accounts
Governance	policy_decisions, ai_decision_traces, human_overrides, kill_switch_events

- Every tenant-owned row includes tenant_id or a database-enforced equivalent scope.
- Provider identifiers require unique constraints scoped to provider/account where appropriate.
- Usage/billing events are append-only after settlement state; corrections use compensating entries.
- High-volume message events should be partition-ready and archival-ready.

101. V2 API Surface

Area	Representative endpoints
Channels	POST /channels/connect, GET /channels/{id}/health, POST /channels/{id}/disconnect
Chatbots	POST /chatbots/generate, POST /chatbots/{id}/test, POST /chatbots/{id}/publish, POST /chatbots/{id}/rollback
Templates	POST /templates/generate, POST /templates/{id}/submit, GET /templates/{id}/status
Campaigns	POST /campaigns, POST /campaigns/{id}/preview, /approve, /schedule, /pause, /resume, /stop
Growth	POST /growth-plans/generate, GET /growth-plans/{id}/recommendations
RCS	POST /rcs/agents/connect, GET /rcs/capabilities/{contact}, POST /rcs/messages
SMS	POST /sms/dlt/profile, POST /sms/templates/validate, POST /sms/messages
Tracking	POST /links, POST /qr-codes, GET /analytics/attribution
Billing	GET /wallet, POST /wallet/topups, GET /usage, GET /rate-cards/effective
Reseller	POST /reseller/tenants, GET /reseller/usage, PUT /reseller/markup

API response contracts must expose normalized domain states, not raw provider response shapes. Preserve raw provider payloads internally for diagnostics.

102. New Domain Events

```
channel.connected
channel.degraded
channel.disconnected
template.generated
template.submitted
template.approved
template.rejected
chatbot.generated
```

```
chatbot.published
campaign.approved
campaign.started
campaign.paused
campaign.completed
recipient.eligible
recipient.suppressed
message.billable
rcs.capability_checked
sms.dlt_validation_failed
wallet.low_balance
wallet.depleted
growth_plan.created
ai.recommendation.created
approval.requested
approval.resolved
```

103. Tenant Product Sitemap — V2

Module	Core screens
Home	Revenue Command Center; AI Actions; Campaign Health; Channel Health
Channels	All Channels; Connect WhatsApp; WhatsApp Health; RCS Agent; SMS/DLT; Provider Settings
Chatbots	All Bots; AI Create Wizard; Visual Builder; Knowledge; Rules; Test Playground; Versions; Analytics
Templates	Template Library; AI Template Studio; Meta Submission; RCS Assets; SMS/DLT Templates; Approval History
Campaigns	All; Create; Audience; Creative; Preview; Schedule; Live Command Center; Experiments; Attribution
AI Growth	Goals; Growth Plan; Marketing Calendar; Recommended Campaigns; Self-Marketing Settings
Inbox	Unified Conversations; AI Handling; Human Handling; Hot Leads; Payment Pending; DND
CRM	Contacts; Leads; Opportunities; Tasks; Appointments; Payments; Imports; Integrations
Tracking	Links; QR Codes; Landing Pages; Forms; Attribution
Analytics	Revenue; Funnel; Channel; Campaign; Template; Agent; Team; Cost & Margin
Billing	Plan; Usage; Wallet; Top-up; Rate Card; Invoices
Settings	Company Brain; Products; Pricing; Offers; Roles; Approvals; Business Hours; Compliance
Reseller (eligible)	Clients; Provisioning; Wallet; Markups; Usage; White-label; Support

104. Tenant Onboarding — “First Value in Under 15 Minutes”

Step	Target UX
1. Create workspace	Company name + industry; AI suggests defaults
2. Import business knowledge	Website/docs/catalogue/CRM; AI summarizes
3. Connect WhatsApp	Embedded/provider wizard + health test
4. Choose objective	Lead generation / sales / support / booking / collections
5. Generate chatbot	AI creates bot + rules + test cases

Step	Target UX
6. Create first template	AI proposes compliant versions; submit for approval where required
7. Import/segment contacts	Eligibility preview before campaign creation
8. Set target	Daily/weekly/monthly result target
9. Preview automation	Simulated messages + costs + handoff rules
10. Launch	Publish bot and campaign; live command center

Onboarding should show blockers early: channel approval, missing consent evidence, missing DLT profile, insufficient wallet, unsupported provider capability. Never wait until “Launch” to reveal a known blocker.

105. Super Admin & Operations Center — V2 Additions

- Provider registry and connector health
- WhatsApp/RCS/SMS account inventory and tenant mapping
- Global template/rejection analytics without leaking tenant content
- Rate-card management and future effective pricing
- Wallet exposure and provider settlement reconciliation
- Reseller management and markup policies
- Global kill switches per provider/channel/campaign type
- AI model/prompt/version rollout controls
- Abuse and unusual-volume detection
- Webhook delay/failure dashboards
- Support diagnostics with permissioned impersonation/read-only view
- Compliance audit exports and data-deletion queues

106. Security, Privacy & Abuse Prevention — V2

Control	Implementation
Credential vault	Encrypt provider tokens/secrets; rotate; never log secret values
Webhook security	Provider signature validation, replay protection, raw-event audit
Tenant isolation	Middleware + repository scope + automated cross-tenant tests + database policies where feasible
Mass-send protection	Role permission, audience preview, approval threshold, rate/frequency guard and kill switch
AI prompt injection	Separate untrusted customer content from system rules; tool/action allowlists
Data minimization	Only retrieve fields necessary for current AI action
Opt-out	Immediate suppression with idempotent update and audit
Exports	Permissioned, logged, rate-limited
Reseller access	Explicit scoping; no cross-client access by default
Cost abuse	Daily spend ceilings, wallet thresholds, anomaly detection, provider-limit alarms

107. Operational SLOs & Scale Targets

Area	Initial target	Scale design
Inbound webhook acknowledgement	< 2 seconds p95	Persist then queue heavy work
AI first response	< 8 seconds p95 where channel permits	Streaming/internal parallel retrieval; bounded retries

Area	Initial target	Scale design
Campaign enqueue	No blocking HTTP loop	Batch creator + Redis queues
Message idempotency	0 duplicate sends from internal retries	Unique idempotency keys
Billing accuracy	100% reconciled billable events after settlement window	Append-only ledger + reconciliation jobs
Channel health detection	< 5 minutes for recurring connector failures	Health jobs + provider webhooks
Tenant isolation incidents	0	Automated security tests and audited access
Campaign emergency stop	Stop new sends within queue-control SLA	Global/tenant/campaign kill switch

108. Fastest V2 Build Plan

Week	Primary deliverable	Parallel dependency track
1	V2 schema, provider/channel adapter contracts, feature flags	Meta/BSP credentials and provider sandbox accounts
2	Rate card, wallet, usage ledger, reseller primitives	RCS/SMS provider contracts and DLT test setup
3	WhatsApp connection wizard + connection state machine	Embedded signup/platform review dependency
4	Template sync + AI Template Studio + submission/status loop	Template approval test cases
5	AI Chatbot Wizard + knowledge ingestion + runtime guardrails	Industry starter packs
6	Chatbot test playground, versions, human handoff	Conversation regression set
7	Bulk campaign builder, eligibility, batching, kill switch	High-volume sandbox/load data
8	Campaign analytics, reply routing, attribution	Payment/CRM event fixtures
9	RCS adapter + capability/fallback + rich composer	RCS agent approval/launch
10	SMS/DLT module + provider adapter + fallback	PE/header/template test credentials
11	AI Growth Planner + self-marketing calendar + experiments	Pilot tenant campaign plan
12	Reseller/white-label, hardening, security/load testing, pilot launch	Operational runbook + support training

Do not delay the first launch waiting for every channel. The commercial MVP should be WhatsApp + AI chatbot + campaign engine + CRM sync + billing/usage + human handoff. Add RCS/SMS behind feature flags as soon as provider onboarding is ready.

109. Definition of Done — Critical V2 Scenarios

PASS WHEN: Tenant connects a WhatsApp account without manually entering a long-lived token; webhook receives inbound message and health shows READY.

PASS WHEN: Tenant describes a business and AI generates a chatbot; owner tests, edits a rule in plain English, publishes, and rolls back to prior version.

PASS WHEN: AI generates three template variants, policy lint catches a risky claim, user fixes/approves, template submits, rejection reason is synced and remediation draft is created.

PASS WHEN: Campaign import contains duplicates, invalid/suppressed contacts and insufficient-variable rows; only eligible recipients enter the send queue.

PASS WHEN: Campaign is paused by kill switch while queued jobs exist; no new outbound messages are executed after the stop boundary.

PASS WHEN: Customer replies to campaign; AI classifies pricing intent, qualifies, books demo, writes CRM activity and routes a high-score lead to a human closer.

PASS WHEN: RCS capability check fails; eligible fallback SMS is sent only when the independent SMS rule permits it.

PASS WHEN: India SMS send is blocked when DLT content-template mapping is missing.

PASS WHEN: Provider cost and tenant sell rate differ; both are recorded and reseller margin is reported correctly.

PASS WHEN: Wallet reaches low-balance threshold; alerts fire, campaign behavior follows tenant rule, and no negative balance occurs unless credit is explicitly enabled.

PASS WHEN: Two tenants use identical phone/contact data; zero data, conversation, campaign or billing leakage occurs.

PASS WHEN: AI recommends a new marketing plan, but external spend/discount action above approval threshold remains pending until authorized.

110. Official Implementation References Verified for V2

Developers must re-check these official sources during implementation because platform rules and commercial models change. Product logic should therefore use configurable policy/rate data rather than assumptions.

Domain	Current implementation reference / note
WhatsApp Business	WhatsApp Business Messaging Policy / Platform policy: opt-in, approved templates for business-initiated messaging where applicable, service-window behavior, automation/human escalation and quality enforcement. https://business.whatsapp.com/policy
Meta developer onboarding	Use current Meta WhatsApp Business Platform/Cloud API and Embedded Signup documentation for the exact authorization, permission, webhook and provider-role requirements. https://developers.facebook.com/docs/whatsapp/
RCS	Google RCS for Business developer guides: agent setup, webhooks, capability checks, rich cards/carousels, suggested replies/actions and billing reports. https://developers.google.com/business-communications/rcs-business-messaging/
India SMS / DLT	TRAI Advice to Senders: Principal Entity registration, header registration, content template and applicable consent requirements. https://www.trai.gov.in/advice-to-senders

111. Developer Non-Negotiables

The application should feel autonomous to the customer, but must be deterministic where mistakes can create legal, financial, account-health or reputational damage.

1. No hard-coded messaging prices, provider limits, tax rates, templates, business hours or discount authority.
2. No AI-generated outbound message bypasses the communication eligibility/policy gate.
3. No bulk-send loop runs synchronously from an HTTP request.
4. No production campaign can be started without an auditable audience snapshot and estimated billable cost.
5. No provider credential is stored in plain text or displayed after capture.
6. No cross-channel fallback assumes consent on the destination channel.
7. No chatbot is allowed to invent pricing, guarantees, legal claims or unavailable product capabilities.
8. No external-platform approval (Meta template, display name, RCS agent etc.) is advertised as guaranteed.
9. No AI optimizer silently changes high-risk rules, spend limits or commercial authority.
10. Every send, reply, AI action, handoff, payment and billable usage event is traceable to tenant and source event.

APPENDIX A — ENHANCED MASTER BUILD PROMPT

For AI coding agents, solution architects, product managers and engineering leads

ROLE & MISSION

Act as a coordinated senior product and engineering team: Principal SaaS Architect, Laravel Backend Lead, React/Next.js Frontend Lead, Flutter Integration Lead, AI/LLM Architect, WhatsApp Business Platform Engineer, RCS Engineer, India SMS/DLT Specialist, CRM Integration Architect, Security/Privacy Engineer, DevOps/SRE Lead, QA Automation Lead, Growth Product Manager, Sales Operations Expert, Revenue Analytics Expert and UX Lead. Your mission is to build a production-grade multi-tenant Autonomous Revenue & Omnichannel Marketing OS that can be sold standalone, embedded inside SFW, white-labeled/resold, or integrated with any CRM.

BUSINESS OUTCOME

The platform must reduce human work across lead capture, consent/eligibility, campaign planning, chatbot setup, messaging, qualification, follow-up, appointment/demo booking, objection handling, payment recovery, reactivation, renewal and reporting. Optimize for qualified revenue and gross margin rather than message volume. “110% target” is a stretch-planning mechanism, not a guaranteed result.

PRIMARY EXPERIENCE

A non-technical customer should be able to sign up, create a company workspace, import business knowledge, connect WhatsApp through the easiest permitted onboarding method, generate a chatbot from website/docs/plain language, create or AI-generate templates, import eligible leads, set a target, preview cost/rules and launch with minimal configuration. All advanced settings remain available for enterprise users.

WHATSAPP ONBOARDING

Implement a connection abstraction supporting Meta direct/Embedded Signup when eligible and approved BSP/provider connectors. Automatically capture account identifiers, set webhooks, sync templates/profile/status, run health diagnostics and expose guided remediation. Never make manual permanent-token pasting the primary onboarding UX. Do not guarantee display-name/logo/template approval.

AI CHATBOT BUILDER

Build a wizard plus visual/natural-language builder. Inputs may be website, documents, product catalogue, CRM fields, industry template or plain-language description. AI creates business summary, intents, lead fields, FAQ answers, branches, CTAs, follow-ups, handoff conditions and test cases. Users can type rules in plain English. Every publish creates a version; support test sandbox, evaluations, rollback, human lock and grounded responses.

AI TEMPLATE STUDIO

Generate multiple WhatsApp template candidates with policy linting, placeholder validation, language localization, CTA suggestions, offer/pricing grounding and quality scoring. Support submission, status sync, rejection reason ingestion, remediation drafts, approval history and immutable versions. Optimize for approval likelihood and conversion but never claim guaranteed Meta approval.

BULK CAMPAIGN ENGINE

Implement audience snapshot, consent/suppression/frequency/template/account-health/budget/schedule/idempotency gates, batching, queue workers, throttling, send-time optimization, A/B tests, kill switches, delivery events, reply routing and revenue attribution. Never implement an unsafe “blast all imported numbers” path.

SELF-MARKETING AUTOPILOT

Allow the customer to set revenue/lead/demo/renewal goals. AI reverse-calculates the funnel, builds a proposed marketing calendar and creates audience, campaign, offer, WhatsApp, RCS, SMS, landing-page, QR/referral and follow-up assets. High-risk changes/spend require approval. Learn from qualified revenue and customer outcomes.

OMNICHANNEL

Use a normalized ChannelAdapter interface. WhatsApp is primary; RCS and SMS are first-class adapters; email/web chat must be extension-ready. Channel routing considers permission, capability, purpose, preference, cost, urgency and engagement. Cross-channel fallback must independently satisfy the destination channel rules.

RCS

Support agent/brand state, webhooks, capability checks, text/media, rich cards, carousels, suggested replies/actions, message expiry, analytics and fallbacks. Use current Google RCS for Business specifications at implementation time.

INDIA SMS & DLT

Build DLT-aware tenant setup with Principal Entity, header, content template, applicable consent-template records, provider mapping and validation. Block sends when required mappings are absent. Use current TRAI/provider rules at implementation time.

CRM & SFW

Implement a universal CRM object model plus adapters. SFW must use SSO and near-real-time two-way sync for contacts, leads, opportunities, tasks, appointments, conversations, payment status and activities. External CRM connectors use mapping, webhooks, reconciliation and idempotency.

RESELLER / WHITE LABEL

Support platform -> reseller -> client tenant hierarchy, client provisioning, reseller wallet/credit, markup, white-label theme/domain, usage reports, margin, scoped APIs and audit access. Keep upstream provider cost private unless configured otherwise.

BILLING

Create effective-dated provider rate cards, channel/category/region slabs, provider cost, platform sell rate, reseller markup, tax rule, subscription fees, wallet/credit, top-ups, usage events, immutable ledgers, low-balance alerts and provider reconciliation. Never hard-code supplied provider commercials.

AI ARCHITECTURE

Prefer one shared orchestration/runtime layer with specialist profiles before creating many independent agent services. Use deterministic tools/actions and structured outputs. Keep AI calls event-driven and budgeted. Store prompt/model/version, retrieved evidence, decision, confidence, tool calls and human override for auditability.

SECURITY

Enforce tenant isolation, RBAC, SSO-ready auth, encrypted provider credentials, webhook verification, replay protection, rate limits, prompt-injection defenses, secrets management, immutable audit trails, data retention controls, opt-out suppression and mass-send approvals/kill switches.

STACK

Fastest production path: Laravel modular monolith API, PostgreSQL, Redis/Horizon queues, React/Next.js web portal, object storage, WebSockets/SSE for live inbox, provider adapters and observability. Do not begin with microservices unless load/organizational boundaries justify them. Keep interfaces clean so domains can be extracted later.

QUALITY & TESTING

Create unit tests for deterministic rules, contract tests for providers, integration tests for webhooks and CRM sync, AI evaluation suites, security/tenant-isolation tests, billing reconciliation tests, load tests for campaigns and golden end-to-end scenarios. A feature is not done until retries, idempotency, failure recovery and audit logs are tested.

DELIVERY METHOD

Break work into epics, schema migrations, API contracts, events, queues, screens, permissions, acceptance criteria and automated tests. Build behind feature flags. Launch WhatsApp + chatbot + campaign + CRM + billing first, then RCS/SMS and self-marketing as provider dependencies clear. Produce runbooks for support, incidents, provider outages, wallet reconciliation and emergency campaign stop.

Master Prompt — Final Directive

Do not simply build features. Build a reliable system of rules, state machines, approvals, event flows, observability and AI assistance that makes revenue operations easy for non-technical businesses while remaining controllable for enterprise customers. If a requirement conflicts with platform policy, privacy, billing correctness, security or tenant isolation, the hard safety/control rule wins and the UI must explain why.